

CHARLES UNIVERSITY

Faculty of Social Sciences

Risk Channels in Monetary Policy

A Quantitative Economics Approach

Master's Thesis

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Declaration

I hereby declare that I have completed this thesis independently using only the listed sources.

In Prague, [date]

Boris Gerát

Abstract

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Keywords

Risk, Monetary policy, BVAR, Financial sector

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Introduction

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Chapter 1

Data and Methodology

This chapter presents the data sources and empirical methodology used in the analysis.

1.1 Data

Data is an essential part of this empirically based thesis. The work leverages time series from multiple sources, and because of this, I divide this section into multiple parts. Each subsection addresses a specific component of the dataset in relation to the corresponding data source used in that part of the analysis. I begin with sectoral-level variables, then proceed to off-the-shelf risk indices, and finally to self-collected data used for constructing custom risk indices.

1.1.1 Financial Sector Variables

The main objective when selecting sectoral-level variables was to identify a set of variables that would be comparable across all sectors. This initially proved difficult due to fragmented data sources, mismatched time windows, and differences in reporting frequencies across the three subsectors.

The banking sector did not pose a significant issue in this regard, as it is one of the most regulated and well-documented sectors. As the primary data source for the banking sector, I use data from the Federal Deposit Insurance Corporation (FDIC), which provides a rich and easily accessible database of balance sheet information for the banking system. Data for this sector was collected first, and the broader availability of variables guided the design and selection of variables for the remaining sectors. While this subsector imposed the fewest constraints on the scope of the research, the final variable set still had to be reduced to ensure comparability with the other two sectors.

For the shadow banking subsector, I use the Z.1 Financial Accounts of the United States, which, to my knowledge, represent one of the most comprehensive sources for this segment of the financial system. Specifically, I use the component labeled “Other financial businesses,” denoted in L.132 which is constructed by subtracting the banking and hedge fund subsectors (reported separately) from the broader financial accounts. This aggregate captures the activity and balance sheet conditions of entities such as money market funds, mutual funds, holding companies, and issuers of asset-backed securities. Initially, I intended to provide two perspectives on the shadow banking sector: one based on the aggregate “Other” category and another focusing on its largest component, money market funds. However, disaggregated data for specific entities were not sufficiently aligned with the structure of the banking sector data. As a result, I focus exclusively on the aggregate representation. An important implication of using this source

is that the analysis is restricted to quarterly frequency, as the Z.1 Financial Accounts are published only on a quarterly basis.

The final subsector considered is the hedge fund sector. The regulatory framework governing the reporting of this subsector is relatively recent, and the data can be found in releases associated with the H.8 financial accounts. The structure of reporting and the available variables are similar to those in the Z.1 accounts. However, due to the recent introduction of these reporting standards, data for hedge funds begin only in 2012 Q4. This represents a substantial limitation compared to the banking and shadow banking sectors, which are available from 1997 Q1. Restricting all sectors to the shorter hedge fund sample would significantly reduce the scope of the analysis. Therefore, I retain the 1997 Q1 – 2024 Q4 window for the banking and shadow banking sectors, while reporting hedge fund results starting from 2012 Q4. Comparisons involving the hedge fund sector must therefore be interpreted with caution, as earlier episodes such as the dot-com bubble or the global financial crisis are not captured in its sample.

Variable	Fraction
Wholesale Funding	$\frac{\text{Brokered Deposits} + \text{Other Borrowed Money}}{\text{Total Assets}}$
Liquidity Buffer	$\frac{\text{Cash} + \text{Treasuries}}{\text{Total Assets}}$
Capital Cushion	$\frac{\text{Total Equity}}{\text{Total Assets}}$
Number of Loans	–

Table 1.1: This constructed balance sheet ratios are used in the main BVAR analysis and serve as the basis for observing impulse response function reactions to increases in risk, which are discussed later in the study.

In Table 1.1, the variables that were consistently collected across all subsectors are presented. The number and variety of these variables allow me to construct ratio-based measures that combine multiple balance sheet components through their interactions. This transformation reduces the dimensionality of the system, which is particularly important given the relatively short time dimension of the dataset used in the BVAR model.

Wholesale funding captures the extent to which a given subsector relies on market-based financing and access to capital markets. The liquidity buffer reflects the share of balance sheet resources allocated to highly liquid assets, providing a measure of short-term resilience to funding shocks. The capital cushion serves as a proxy for leverage, indicating the degree of risk-taking behavior within the subsector. The number of loans as they are as it carries substantial informational content on its own and directly captures credit provision dynamics.

This broader set of variables is designed to capture the overall behavior of each subsector when subject to monetary policy and risk shocks. At the same time, the selection enables internal consistency

checks and robustness in the analysis of impulse response functions. For instance, changes in leverage should be accompanied by corresponding movements in wholesale funding, which provides an additional layer of validation for the identified transmission mechanisms.

1.1.2 Monetary Policy Variable

For the role of the main monetary policy variable, I considered multiple candidates and approaches. However, for the final analysis, I choose to use the shadow rate proposed by Wu and Xia (2016). This measure provides a more accurate representation of the stance of monetary policy when the effective federal funds rate is constrained by the zero lower bound.

During the Global Financial Crisis, monetary policy became highly accommodative and relied on non-standard tools such as quantitative easing alongside conventional rate setting. These measures are not fully captured by the standard effective federal funds rate, as nominal interest rates are bounded at zero and cannot reflect additional monetary accommodation. The Wu and Xia shadow rate addresses this limitation by allowing for negative values, thereby capturing the implicit policy stance under unconventional monetary policy. At the same time, it closely tracks the effective federal funds rate during periods away from the lower bound, ensuring consistency across regimes.

This methodology is particularly useful during and immediately after the 2008 financial crisis. However, once the shadow rate converges back to the official effective federal funds rate in more stable periods, the difference between the two becomes negligible. As the shadow rate series is not available after 2023, and re-estimating the model for a relatively short extension of the sample is not essential for the purposes of this analysis, I replace the missing observations with the effective federal funds rate. Given the close co-movement of the two series outside of the zero-lower-bound period, this substitution does not materially affect the results.

Initially, I also considered the methodology of Romer and Romer (2004), which isolates unexpected monetary policy shocks by removing the anticipated component of policy changes. This approach is useful in settings where expectations may bias the estimated effects of monetary policy, for example through the so-called price puzzle. However, given the lower frequency of the data used in this analysis, such issues are less pronounced. In this context, the shadow rate approach provides a more suitable measure of the overall stance of monetary policy and is therefore preferred.

1.1.3 Benchmark Risk Indices

To measure effects of my custom risk indices I am introducing large quantity of benchmark risk indices both from traditional time series based kinds and from sentiment ones too. In this way even if the custom risk indices will fail to deliver effect the BVAR analysis that relies on them will still be valuable. The used risk indices can be listed in the table below with their sources used in analysis and results and short explanation.

Table 1.2: Financial Risk and Uncertainty Indices

Abbreviation	Full Name	Description
STLFSI	St. Louis Fed Financial Stress Index	Composite index constructed from 18 financial market variables, including interest rate spreads, volatility measures, and yield curve indicators.
NFCI	Chicago Fed National Financial Conditions Index	Broad index based on over 100 financial indicators capturing risk, credit, and leverage conditions in the U.S. financial system.
KCFSI	Kansas City Financial Stress Index	Stress index constructed from 11 financial variables, focusing on yield spreads and market volatility across financial sectors.
VIX	CBOE Volatility Index	Market-based measure of expected volatility derived from S&P 500 option prices; commonly interpreted as a forward-looking risk indicator.
EPUI	Economic Policy Uncertainty Index	News-based index measuring policy-related economic uncertainty using frequency of newspaper articles combined with policy indicators.
NSI	News Sentiment Index (Shapiro et al.)	Text-based sentiment index constructed from news articles, capturing positive and negative economic tone across a large corpus of media sources.

Note: The indices combine financial market data and textual information to capture different dimensions of risk and uncertainty relevant for macro-financial analysis.

In the majority of related studies, the STLFSI (often referred to simply as the Financial Stress Index, FSI) serves as a primary proxy for financial risk, typically alongside the VIX index. Together, these two measures provide a robust framework for analyzing risk responses. The STLFSI captures broad financial stress conditions, while the VIX represents a purely market-based measure derived from S&P 500 option prices, reflecting forward-looking expectations of volatility.

However, in this analysis, I extend the set of risk indicators by incorporating additional indices constructed by Federal Reserve Banks beyond the St. Louis Fed. In particular, I include the NFCI and KCFSI, which differ in their construction methodologies relative to the STLFSI. The NFCI can be viewed as a higher-dimensional measure, incorporating a large number of financial variables to capture a broader range of risk, credit, and leverage conditions. In contrast, the KCFSI adopts a more parsimonious approach, focusing on a smaller set of variables closely linked to financial market volatility and stress dynamics.

In addition to financial stress indices, I incorporate sentiment-based measures of risk. The Economic

Policy Uncertainty Index (EPUI) is included as it is explicitly designed to capture uncertainty in the economic and policy environment through a structured, news-based methodology. In contrast, the Shapiro News Sentiment Index serves a more general purpose, capturing overall economic tone in news coverage. As such, while it provides valuable information, it does not exclusively target risk or uncertainty in the same way as EPUI.

Overall, this set of risk indices provides a broad and complementary representation of financial risk and uncertainty. It enables a more comprehensive assessment of the risk channel and facilitates robustness checks within the BVAR framework, particularly when compared to the custom risk indices constructed in this study.

1.1.4 Custom Risk Indices

In this thesis I am introducing three types of risk indices. First type is time series based similar to the risk indices created by federal reserve banks. Second type are risk indices are based on textual data which is then transformed into sentiment similarly as Shapiro news index and EPU index. The last type is the combination of them, to try leverage signals from both.

Time Series Based Indices

To construct the custom risk index, I follow the general methodology employed by the Federal Reserve Bank of St. Louis in the development of financial stress indices. This approach is based on selecting variables that exhibit a strong signal with respect to financial risk and extracting their common component through principal component analysis.

In this thesis, I extend this methodology by introducing a hierarchical factor structure. Specifically, I first group variables according to the type of signal they capture, ranging from policy-related indicators to market-based measures. Within each group, I extract the first principal component, which represents the dominant common movement specific to that category of variables.

In the second stage, I take the first principal component of these group-level factors. This results in a final aggregate factor that captures the common variation across all signal types. In this way, the constructed risk index reflects a broad and integrated measure of financial risk, incorporating information from multiple dimensions of the financial system.

All variables included in the construction of the index are listed below.

Table 1.3: Variables Used in Hierarchical Risk Factor Construction

Category	Variables
Credit	Moody's Baa Corporate Bond Spread, ICE BofA US High Yield Spread, ICE BofA US Corporate Bond Spread
Funding	Commercial Paper Spread, Prime Rate Minus Federal Funds Rate, Commercial Paper Minus 3-Month Treasury Spread
Policy	10Y-2Y Treasury Spread, 10Y-3M Treasury Spread, 10Y Treasury Minus Federal Funds Rate
Market	NASDAQ Composite Index (log returns), S&P 500 Index (log returns), Russell Index (log returns)

Note: Variables are grouped according to their economic interpretation into credit risk, funding conditions, monetary policy stance, and market-based indicators. Within each group, the first principal component is extracted and subsequently aggregated into a final risk factor.

The choice of variables was straightforward, as the objective was to achieve results comparable to existing Federal Reserve risk indices. This ensures that the constructed index remains methodologically comparable to established benchmarks, while still distinguishing itself through its specific variable selection and the use of hierarchical factor extraction.

Sentiment Based Indecies

In this section I want to introduce textual based data that were collected from relevant sources that should carry information about risk and confidence of the US economy. All collection was done by me using legal web scraping methods. The selection of textual sentiment sources can be seen in figure below.

Main Sentiment Data Sources

Number of Unique Observations by Source

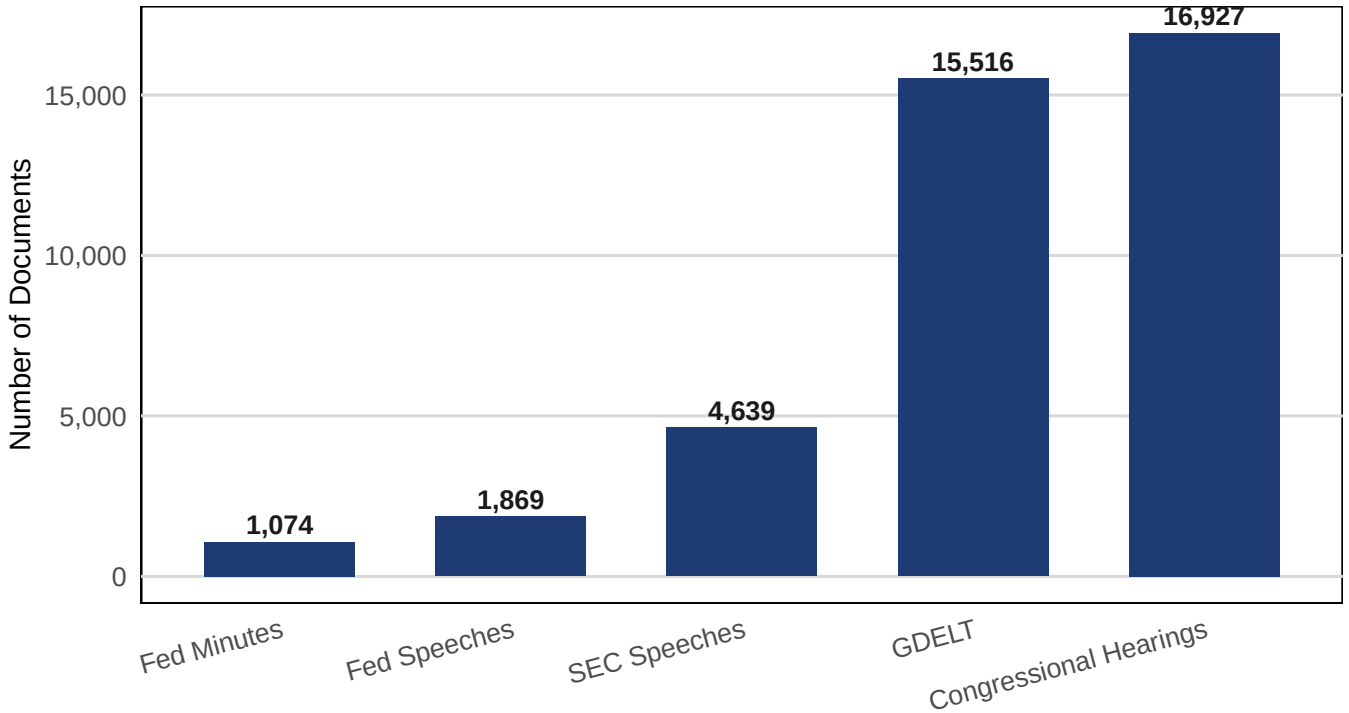


Figure 1.1: Main Sentiment Data Sources

As can be seen, the sentiment index is constructed from a diverse set of textual data sources. The largest contribution comes from congressional hearings, filtered to include only economically relevant committees. The second-largest source is the GDELT project, which represents one of the most extensive collections of global news events. While GDELT does not provide direct access to the full textual content, it offers pre-constructed indices derived from news coverage. In this analysis, only economically relevant entries are selected.

The remaining sources are smaller in volume but provide highly relevant economic information. These include Securities and Exchange Commission (SEC) speeches, Federal Reserve minutes, and Federal Reserve speeches. Although these datasets are more limited in size, they are closely aligned with economic policy and institutional communication, and therefore contain strong informational content for sentiment extraction.

Overall, the dataset is more heavily weighted toward policy-based sources of sentiment. However, the inclusion of GDELT news data helps balance this by incorporating broader media-based information. Initially, I intended to further expand the dataset by including textual data from platforms such as Reddit and X (formerly Twitter). However, access to X data requires paid subscriptions, which are beyond the scope of a master's research budget. In the case of Reddit, although large historical datasets are available through archived sources, the data collection process proved to be computationally intensive given the available hardware constraints. Despite these limitations, such data sources represent a promising extension and could further enhance the sentiment analysis.

The final sentiment measure combines this text-based index with the time-series-based risk indices. This is achieved through the extraction of their first principal component, consistent with the methodology applied throughout this thesis. In this way, the resulting index captures both high-frequency textual

sentiment and broader macro-financial dynamics.

1.2 Methodology

In this section, I present the methodological framework and identification strategies used in this thesis. I begin with factor models for dimensionality reduction, then proceed to sentiment extraction methods, and conclude with the specification of the Bayesian VAR (BVAR) model.

1.2.1 Expectation Maximization Principal Component Analysis (EMPCA)

Based on prior experience with similar research problems, I choose to use the EMPCA model. This approach is both simple and powerful. Its main advantage lies in its ability to handle missing data through the expectation-maximization procedure, while maintaining a linear and fully interpretable structure. Unlike more complex nonlinear methods, EMPCA preserves transparency in the estimation process, which is desirable in an empirical setting.

In this thesis, EMPCA is used whenever extraction of the first principal component is required, forming a core building block of the hierarchical factor construction. The EMPCA follows original methodology proposed by S. Roweis (1998).

Alternative approaches were also considered. In particular, the NIPALS algorithm and Dynamic Factor Models (DFM) were evaluated. However, NIPALS either underperformed relative to EMPCA or produced very similar results without additional benefits. Dynamic Factor Models, on the other hand, encountered convergence issues when applied to datasets with low dimensionality and relatively short time series. For these reasons, EMPCA is selected as the primary method throughout the thesis.

EMPCA Model Specification

The EMPCA framework assumes that the observed data matrix $X \in \mathbb{R}^{T \times N}$ can be represented by a low-dimensional factor structure:

$$X = WZ + \varepsilon \quad (1.1)$$

where $W \in \mathbb{R}^{T \times k}$ contains the latent factors, $Z \in \mathbb{R}^{k \times N}$ are the factor loadings, and ε is an idiosyncratic error term.

The estimation proceeds via the Expectation-Maximization algorithm.

E-step (Expectation) Given current estimates of loadings $Z^{(t)}$, the latent factors are updated as:

$$W^{(t+1)} = XZ^{(t)\top} (Z^{(t)}Z^{(t)\top})^{-1} \quad (1.2)$$

Step interpretation: This step estimates the latent factors W by projecting the observed data onto the current loading structure.

M-step (Maximization) Given updated factors $W^{(t+1)}$, the loadings are updated as:

$$Z^{(t+1)} = (W^{(t+1)\top} W^{(t+1)})^{-1} W^{(t+1)\top} X \quad (1.3)$$

Step interpretation: This step updates the factor loadings Z to best fit the observed data given the estimated latent factors.

Objective Function The algorithm iteratively minimizes the reconstruction error:

$$\min_{W, Z} \|X - WZ\|_F^2 \quad (1.4)$$

Step interpretation: The goal is to find the factor representation that best approximates the observed data in a least-squares sense.

Handling Missing Data Let M denote a masking matrix indicating observed entries. The objective becomes:

$$\min_{W, Z} \|M \odot (X - WZ)\|_F^2 \quad (1.5)$$

Step interpretation: Only observed values contribute to the estimation, allowing EMPCA to naturally handle missing data without prior imputation.

1.2.2 Granger Causality Analysis

In this thesis, I employ Granger causality tests to examine whether risk indices contain predictive information about the dynamics of financial sector subsectors. This approach provides a simple yet informative framework for assessing the direction of information flow and the propagation of risk across different segments of the financial system. In particular, it allows me to test whether past values of risk indices improve the prediction of sectoral variables beyond their own lagged values.

Formally, the following regression is estimated:

$$y_t = \alpha + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \beta_j x_{t-j} + \varepsilon_t \quad (1.6)$$

where y_t represents the variable of interest (e.g., a financial sector indicator), and x_t denotes the risk index. The null hypothesis of no Granger causality is given by:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_q = 0 \quad (1.7)$$

Rejection of the null implies that past values of the risk index provide statistically significant information for predicting y_t , indicating a directional relationship consistent with risk transmission.

This methodology follows the standard Granger causality framework introduced by Granger (1969)

1.2.3 Text-to-Sentiment Models

To ensure diversity in both methodology and results, I employ two distinct sentiment extraction approaches. The first is the well-established lexicon-based VADER model, while the second is the more

recent FINBERT model, which is specifically designed for financial and economic text analysis.

VADER is a lexicon-based sentiment analysis model developed by Hutto and Gilbert (2014), primarily designed for short, informal textual data. It relies on a predefined dictionary of words with associated sentiment intensity scores ranging from +4 (strongly positive) to -4 (strongly negative). These word-level scores are aggregated into a normalized sentiment measure in the range $[-1, 1]$. The main advantage of VADER lies in its transparency and interpretability, as the scoring mechanism is fully traceable and does not rely on complex model training.

However, VADER is not specifically tailored to financial contexts and may therefore miss important domain-specific nuances. In addition, its performance may deteriorate when applied to longer and more structured texts, such as congressional hearings, which constitute a substantial portion of the dataset. Despite these limitations, its robustness and interpretability make it a valuable benchmark model in the analysis.

FINBERT, in contrast, represents a modern approach to sentiment analysis in financial applications. It is based on the Bidirectional Encoder Representations from Transformers (BERT) model introduced by Devlin et al. (2019), which utilizes a transformer-based neural network architecture. Unlike lexicon-based methods, BERT models evaluate words in context, allowing for a more accurate representation of meaning within a sentence. FINBERT, as developed by Araci (2019), is a version of BERT that has been fine-tuned specifically on financial text, making it particularly well-suited for economic applications.

Using both VADER and FINBERT allows for a comparison between a transparent, rule-based benchmark and a context-aware neural model. This dual approach enhances the robustness of the sentiment analysis and enables an evaluation of how model choice influences the resulting sentiment measures.

For implementation, I use the Python `vaderSentiment` library for VADER and the Hugging Face `transformers` framework for FINBERT. The resulting sentiment scores are subsequently normalized and used as inputs into the factor models, where they contribute to the construction of custom risk indices.

1.2.4 Bayesian VAR

In this thesis, I employ a Bayesian Vector Autoregression (BVAR) model to analyze the dynamic interactions between monetary policy, risk indices, and financial sector variables. The Bayesian framework is particularly suitable in this context, as it enables reliable estimation in settings with relatively short time series by incorporating prior information and controlling for overparameterization. This approach follows the seminal contributions of Doan, Litterman, and Sims (1984) and Litterman (1986), with further extensions in hierarchical prior selection as proposed by Giannone, Lenza, and Primiceri (2015).

Specifically, I use a hierarchical BVAR specification, in which hyperparameters governing the prior distribution are estimated from the data. This allows the model to determine the appropriate degree of shrinkage in a data-driven manner, improving both stability and predictive performance.

Model Specification

The reduced-form VAR(p) model is given by:

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \varepsilon_t, \quad (1.8)$$

where $y_t \in \mathbb{R}^n$ is a vector of endogenous variables, A_i are coefficient matrices, and $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ is a vector of reduced-form shocks.

The model can be written in compact matrix form as:

$$Y = XB + E, \quad (1.9)$$

where Y is the matrix of observations, X contains lagged values of the endogenous variables, and B is the matrix of coefficients.

Within the Bayesian framework, prior distributions are imposed on the parameters:

$$B \sim \mathcal{N}(B_0, V_0), \quad \Sigma \sim \mathcal{IW}(S_0, \nu_0), \quad (1.10)$$

where $\mathcal{IW}$ denotes the inverse-Wishart distribution. These priors introduce shrinkage towards a structured baseline, typically associated with Minnesota-type priors, which reduce estimation uncertainty in small samples.

The posterior distribution is obtained using Bayes' rule:

$$p(B, \Sigma \mid Y) \propto p(Y \mid B, \Sigma) p(B, \Sigma), \quad (1.11)$$

which combines prior information with the likelihood implied by the observed data.

In the hierarchical extension, hyperparameters governing the prior distribution are themselves treated as random variables:

$$\lambda \sim p(\lambda), \quad B \mid \lambda \sim \mathcal{N}(B_0(\lambda), V_0(\lambda)). \quad (1.12)$$

This additional layer allows the degree of shrinkage to be determined endogenously, which is particularly advantageous when working with limited time series length.

Identification via Cholesky Decomposition

To identify structural shocks, I employ a recursive identification scheme based on the Cholesky decomposition of the covariance matrix:

$$\Sigma = PP^\top, \quad (1.13)$$

where P is a lower triangular matrix.

The structural form can be written as:

$$P^{-1}\varepsilon_t = u_t, \quad (1.14)$$

where u_t represents orthogonal structural shocks.

The ordering of variables plays a crucial role in this identification. Let the vector of endogenous variables be defined as:

$$y_t = \begin{bmatrix} MP_t \\ Risk_t \\ Sector_t \end{bmatrix}, \quad (1.15)$$

where MP_t denotes the monetary policy variable, $Risk_t$ the risk index, and $Sector_t$ the financial sector variable.

Under this ordering, the Cholesky decomposition imposes the following structure:

$$P = \begin{bmatrix} p_{11} & 0 & 0 \\ p_{21} & p_{22} & 0 \\ p_{31} & p_{32} & p_{33} \end{bmatrix}, \quad (1.16)$$

which implies that monetary policy shocks can contemporaneously affect both risk and sectoral variables, while risk can affect sectoral variables within the same period, but not vice versa.

An alternative specification reverses the ordering of the first two variables:

$$y_t = \begin{bmatrix} Risk_t \\ MP_t \\ Sector_t \end{bmatrix}, \quad (1.17)$$

which corresponds to a different decomposition matrix:

$$P = \begin{bmatrix} p_{11} & 0 & 0 \\ p_{21} & p_{22} & 0 \\ p_{31} & p_{32} & p_{33} \end{bmatrix}, \quad (1.18)$$

but now with risk ordered first, allowing risk shocks to contemporaneously affect monetary policy.

Both specifications are estimated to assess the robustness of the results with respect to identification assumptions. Alternative identification strategies based on sign restrictions were also considered, but did not yield substantially different impulse responses. This outcome is likely driven by the relatively short time series and the use of Minnesota-type priors, which impose substantial shrinkage and reduce the range of admissible parameter configurations.

Chapter 2

Results

In this chapter, I present the results of this thesis across all models and estimations. I begin with the risk indices and their dynamics, documenting how each measure evolves over the sample period. This is followed by the results from the Bayesian vector autoregressions and an analysis of risk shock propagation across the economy.

2.1 Custom risk index dynamics

In this subsection, I take a closer look at the performance of the custom risk indices relative to the included benchmarks. The results are presented primarily through graphical analysis and a correlation table. Overall, this part of the thesis sheds light on both the benefits and difficulties of constructing indices from scratch, as well as the tradeoffs involved in incorporating sentiment proxies.

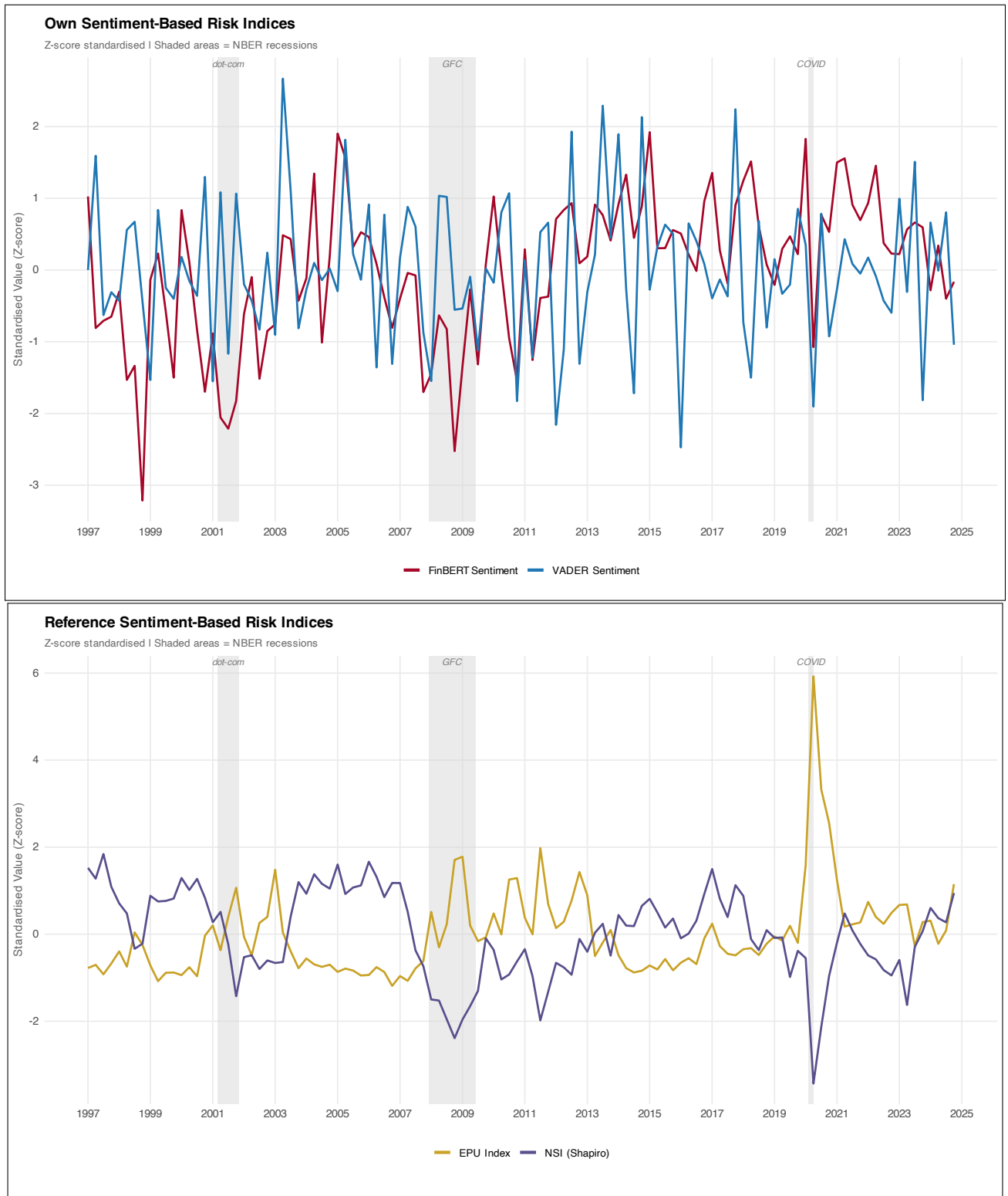


Figure 2.1: INSERT CAPTION HERE.

The estimated custom sentiment indices and their benchmark counterparts are presented in Figure 2.1. The upper panel displays the dynamics of the custom risk indices estimated using the VADER and FinBERT models, while the lower panel shows the reference benchmark indices.

The behaviour of both custom indices is markedly volatile throughout the estimated sample. Focusing specifically on the VADER sentiment index, the series does not reliably spike during major recessions in the way one would expect from a well-performing risk indicator. The index fails to respond consistently to the dot-com bubble and the Global Financial Crisis, and only during the COVID-19 pandemic does it move in the expected direction, correctly declining in line with the NBER recession definition.

The FinBERT sentiment index, by contrast, exhibits considerably better performance. It displays lower overall volatility and more consistent countercyclical behaviour, correctly capturing all three major recession episodes. The most pronounced responses are observed during the dot-com bubble and the Global Financial Crisis, during which the index declines sharply and persistently, suggesting that FinBERT’s domain-specific financial language training translates into more economically meaningful sentiment extraction.

Cross-checking these results against the benchmark risk indices reveals that the reference series exhibit substantially lower volatility during non-recessionary periods, which is a desirable property for a risk indicator. The Shapiro News Sentiment Index is notably less volatile and less reactionary compared to the Economic Policy Uncertainty Index, yet both indices appear to detect signals broadly consistent with those captured by the custom FinBERT index, lending some external validity to the latter’s construction.

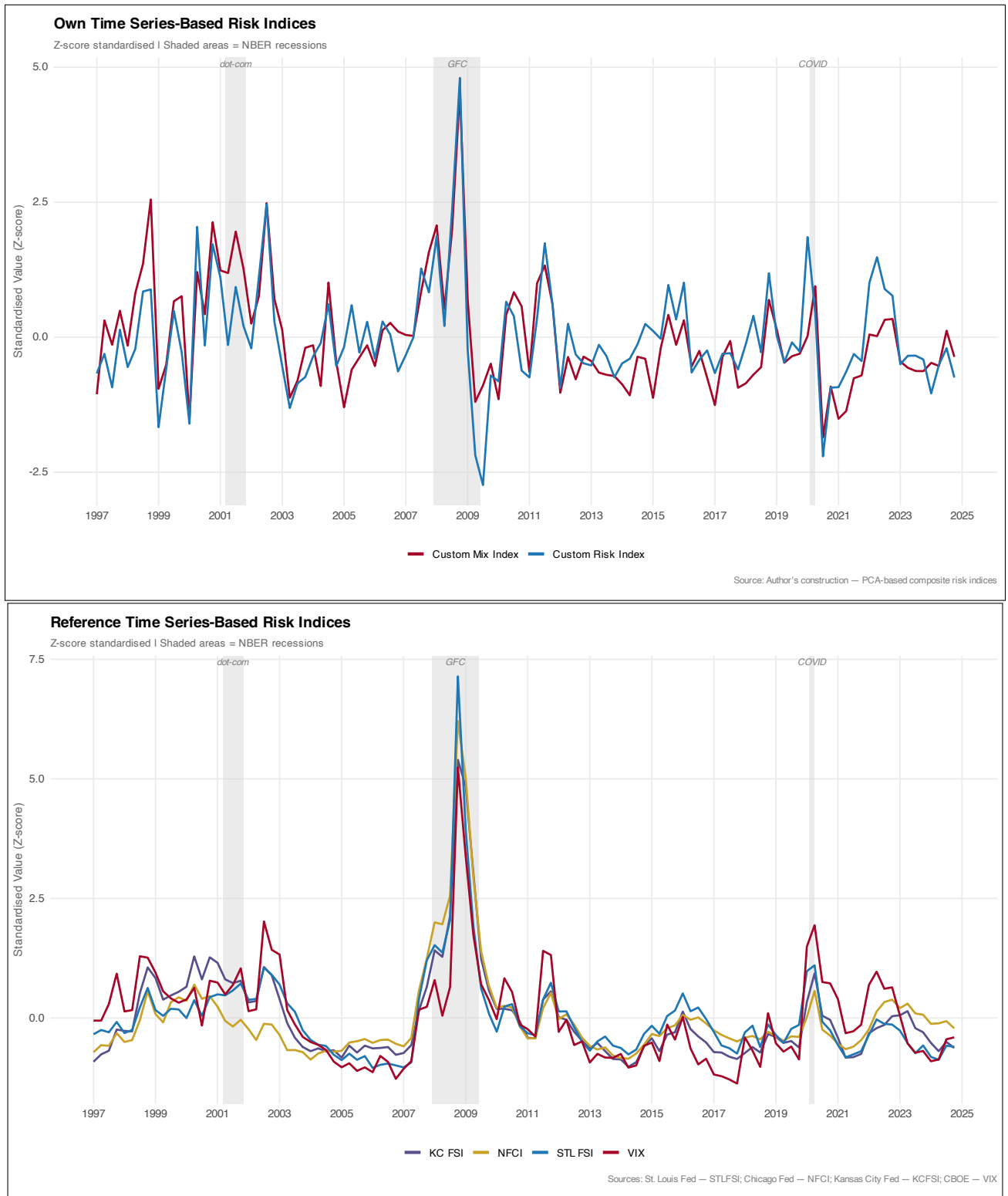


Figure 2.2: INSERT CAPTION HERE.

The results for the time series-based risk indices are presented in Figure 2.2. A broadly similar dynamic emerges as with the sentiment-based indices discussed above. The main recessionary signal appears to be captured, however both custom indices exhibit a tendency to absorb excess noise during tranquil periods. A particularly important finding in this section is that the inclusion of sentiment appears to stabilise the custom time series-based index, rendering it more comparable to the reference benchmark indices.

More specifically, the incorporation of sentiment into the composite index enhances its responsiveness around major economic events, amplifying spikes in a way that the purely time series-based index does

not. This is most evident during the dot-com bubble, where the sentiment-augmented index reaches notably higher levels compared to the benchmark indices, a plausible and economically meaningful signal that the reference indices fail to fully capture. This represents perhaps the most distinctive contribution of the sentiment component to the overall index dynamics.

Furthermore, sentiment inclusion appears to stabilise the post-crisis recovery path following the Global Financial Crisis. Unlike the classic time series index, which remains depressed for an extended period, the sentiment-augmented version converges more smoothly toward baseline levels, bringing its behaviour closer in line with the reference indices. This suggests that sentiment carries forward-looking information that helps the index recover in a more timely and economically coherent manner.

During the COVID-19 pandemic, both custom indices, with and without sentiment, decline considerably more sharply than the benchmark indices. This points to a potential limitation of the constructed indices, namely an oversensitivity to acute crisis episodes, which may warrant further investigation in the robustness analysis.

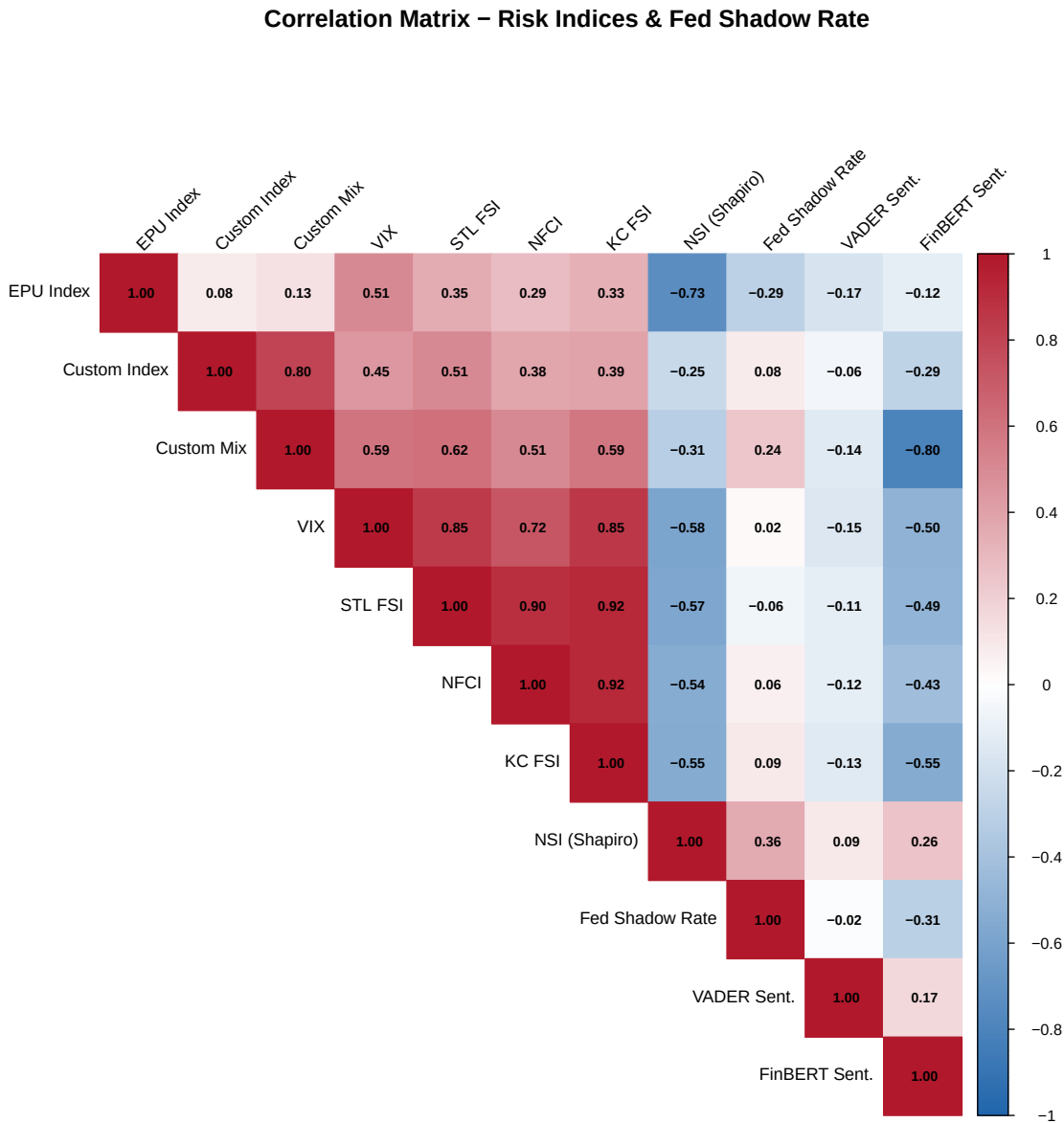


Figure 2.3: INSERT CAPTION HERE.

Figure 2.3 presents a correlation matrix of all risk indices used in this analysis. As expected, the time-series-based risk indices exhibit higher correlations with the Fed benchmark indices compared to the sentiment-based ones. Among the sentiment indices, the Shapiro news sentiment index and the custom FinBERT index appear negatively correlated with the broader risk indices, while the custom VADER sentiment index shows no meaningful correlation with any of the other indices. The EPU index, due to its construction methodology, displays a slight positive correlation with the main Fed risk indices but little relationship with the custom risk indices, suggesting it captures a different type of uncertainty signal, one that appears more aligned with the Shapiro news index than with the financial stress measures.

Interestingly, the three indices most correlated with the Fed shadow rate are the EPU index (negatively), the custom mixed index, and the Shapiro news index. This is a noteworthy finding, as all three carry sentiment-related information, potentially highlighting the role of sentiment dynamics in the transmission of monetary policy shocks.

The following subsection presents results from Granger causality tests, which shed further light on the predictive capacity of the risk indices beyond graphical representations and simple correlation analysis.

2.2 Granger Causality Results

In this subsection, I present the results from Granger causality analysis, which sheds light on the relationships between risk indices and subsectors of the financial sector. This analysis allows the study to examine not only the predictive power of risk indices for subsectors, but also the reverse relationship, as well as the lag length relevant for predictive power. All results from the Granger causality analysis are presented in Table 2.1 below.

Table 2.1: Granger Causality Results Across Financial Sectors

Sector	Risk Index	Risk $\rightarrow$ Factor	Factor $\rightarrow$ Risk	Lags	Causality Direction
Banking	STLFSI	$p=0.0000$ ***	$p=0.9477$ n.s.	4	STLFSI $\rightarrow$ Banking
	NFCI	$p=0.0000$ ***	$p=0.9148$ n.s.	4	NFCI $\rightarrow$ Banking
	KCFSI	$p=0.0000$ ***	$p=0.9677$ n.s.	4	KCFSI $\rightarrow$ Banking
	VIX	$p=0.0000$ ***	$p=0.9237$ n.s.	4	VIX $\rightarrow$ Banking
	EPUI	$p=0.0004$ ***	$p=0.2411$ n.s.	4	EPUI $\rightarrow$ Banking
	Custom Index	$p=0.0004$ ***	$p=0.7594$ n.s.	4	Custom Index $\rightarrow$ Banking
	Custom Mix	$p=0.0172$ **	$p=0.2398$ n.s.	4	Custom Mix $\rightarrow$ Banking
	NSI (Shapiro)	$p=0.0895$ *	$p=0.0987$ *	4	No causality
	Sentiment FinBERT	$p=0.2202$ n.s.	$p=0.1919$ n.s.	4	No causality
	Sentiment VADER	$p=0.2767$ n.s.	$p=0.2213$ n.s.	4	No causality
	Shadow Rate	$p=0.6486$ n.s.	$p=0.4902$ n.s.	4	No causality
Shadow Banking (Small)	NFCI	$p=0.7110$ n.s.	$p=0.0003$ ***	4	Shadow Small $\rightarrow$ NFCI
	Custom Index	$p=0.0153$ **	$p=0.0019$ ***	4	Bidirectional
	STLFSI	$p=0.7120$ n.s.	$p=0.0021$ ***	4	Shadow Small $\rightarrow$ STLFSI
	KCFSI	$p=0.3644$ n.s.	$p=0.0045$ ***	4	Shadow Small $\rightarrow$ KCFSI
	Custom Mix	$p=0.1301$ n.s.	$p=0.0116$ **	4	Shadow Small $\rightarrow$ Custom Mix
	NSI (Shapiro)	$p=0.0325$ **	$p=0.2018$ n.s.	4	NSI $\rightarrow$ Shadow Small
	Sentiment VADER	$p=0.0955$ *	$p=0.7962$ n.s.	4	No causality
	EPUI	$p=0.1394$ n.s.	$p=0.4509$ n.s.	4	No causality
	VIX	$p=0.7368$ n.s.	$p=0.1453$ n.s.	4	No causality
	Sentiment FinBERT	$p=0.8220$ n.s.	$p=0.2069$ n.s.	4	No causality
	Shadow Rate	$p=0.2317$ n.s.	$p=0.4786$ n.s.	4	No causality
Shadow Banking (Broad)	NSI (Shapiro)	$p=0.1769$ n.s.	$p=0.0000$ ***	4	Shadow Broad $\rightarrow$ NSI
	Sentiment FinBERT	$p=0.9165$ n.s.	$p=0.0000$ ***	4	Shadow Broad $\rightarrow$ FinBERT
	Custom Mix	$p=0.8181$ n.s.	$p=0.0001$ ***	4	Shadow Broad $\rightarrow$ Custom Mix
	NFCI	$p=0.5936$ n.s.	$p=0.0015$ ***	4	Shadow Broad $\rightarrow$ NFCI
	STLFSI	$p=0.4723$ n.s.	$p=0.0071$ ***	4	Shadow Broad $\rightarrow$ STLFSI
	KCFSI	$p=0.5409$ n.s.	$p=0.0225$ **	4	Shadow Broad $\rightarrow$ KCFSI
	EPUI	$p=0.0438$ **	$p=0.0334$ **	4	Bidirectional
	Custom Index	$p=0.3497$ n.s.	$p=0.0424$ **	4	Shadow Broad $\rightarrow$ Custom Index
	Sentiment VADER	$p=0.0555$ *	$p=0.6876$ n.s.	4	No causality
	VIX	$p=0.6093$ n.s.	$p=0.2896$ n.s.	4	No causality
	Shadow Rate	$p=0.4932$ n.s.	$p=0.6206$ n.s.	4	No causality
Hedge Fund	NSI (Shapiro)	$p=0.2374$ n.s.	$p=0.0167$ **	4	Hedge Fund $\rightarrow$ NSI
	Sentiment VADER	$p=0.2235$ n.s.	$p=0.0704$ *	4	No causality
	Custom Mix	$p=0.1166$ n.s.	$p=0.8107$ n.s.	4	No causality
	Sentiment FinBERT	$p=0.1538$ n.s.	$p=0.7999$ n.s.	4	No causality
	EPUI	$p=0.3193$ n.s.	$p=0.9367$ n.s.	4	No causality
	STLFSI	$p=0.4158$ n.s.	$p=0.9771$ n.s.	4	No causality
	VIX	$p=0.4302$ n.s.	$p=0.9784$ n.s.	4	No causality
	NFCI	$p=0.4339$ n.s.	$p=0.8751$ n.s.	4	No causality
	Custom Index	$p=0.4744$ n.s.	$p=0.8527$ n.s.	4	No causality
	Shadow Rate	$p=0.5889$ n.s.	$p=0.9492$ n.s.	4	No causality
	KCFSI	$p=0.6621$ n.s.	$p=0.8231$ n.s.	4	No causality

Notes: Significance: *** $p < 0.001$, ** $p < 0.05$, * $p < 0.1$, n.s. not significant. All tests use 4 lags (BIC-selected).

Rows sorted by strength of dominant causality direction within each sector. Shadow Rate = Wu-Xia shadow federal funds rate.

The inclusion of custom risk indices alongside a broad set of benchmark indices pays off clearly in this subsection of results. The wide range of risk signal carriers exhibits heterogeneous predictive performance across the financial sector subsectors.

The banking sector, which opens the table, behaves as expected: it is predominantly a risk receiver rather than a risk creator or amplifier. Predictive power flows in one direction only, from the risk indices toward the banking sector. The strongest predictive performance comes from the benchmark Fed risk indices FSI, NFCI, and KCFSI. The VIX, a purely market-based index, performs on par with these Fed measures, which further underlines its widespread use in both academia and industry. A notable contribution also comes from EPUI, the only sentiment-based index that performs well in this context, likely reflecting the distinct methodology underlying its construction. Both custom risk indices show strong predictive power toward the banking subsector, with the purely time-series-based index achieving a lower p-value. The custom sentiment indices, however, do not appear capable of predicting banking sector dynamics, and the same holds for the Shapiro news index. The shadow rate itself shows no meaningful predictive power in either direction, and the reverse relationship, in which the banking subsector predicts risk indices, is similarly absent. Only a faint signal appears in relation to the Shapiro news index.

The picture changes drastically when attention shifts to the shadow banking core subsector. The direction of the relationship appears to flip: the shadow banking sector emerges as the driver of predictive power rather than its recipient. This pattern holds most clearly for the classic benchmark Fed indices and for both the custom time-series risk index and the custom mixed index. Notably, the custom time-series risk index exhibits bidirectional predictive power, a property not shared by any of the other risk indices considered. The sentiment-based risk indices do not display a clean relationship with this subsector. The exception is the Shapiro news index, which shows predictive power over the shadow banking core. The shadow rate contributes negligible predictive power in either direction.

Results do not change dramatically when the broader shadow banking factor is examined. The predictive power originating from the subsector becomes even more pronounced and extends to a wider set of risk indices, including the Shapiro news index and the custom FinBERT sentiment index. In this case, the only index displaying bidirectional predictive power is EPUI, though the significance is not strong. The comparison between the core and broad shadow banking factors nonetheless points to a consistent and clear directional shift relative to the banking sector. The shadow rate exhibits essentially the same negligible impact observed for the core subsector.

The table concludes with the hedge fund subsector, where predictive power is considerably weaker overall. Only marginally significant results emerge, and these are concentrated among the sentiment indices, specifically the Shapiro news index and the custom VADER sentiment index. In both cases, the direction of predictive power runs from the hedge fund subsector toward the risk indices rather than the reverse, suggesting that hedge fund dynamics may carry some information content for sentiment-based risk measures, even if the overall signal is modest.

2.3 BVAR results

In this section I present results from hierarchical Bayesian vector autoregressions. Due to the number of individual risk indices considered, the full set of results cannot be presented in the main body of the thesis. I have therefore selected only the most representative indices for inclusion here, with the remaining results alongside the residual correlation matrices from each BVAR model displayed in the appendix.

The results are organized by risk index, with each subsection covering one index. Within each

subsection, I first present the impulse response function of the risk index to a shadow rate shock, followed by the responses of the sectoral variables across all three financial subsectors. This structure allows the reader to easily grasp the results and compare transmission dynamics across subsectors within a given risk environment.

The inclusion of both established reference indices and custom indices serves an additional purpose as a built-in robustness mechanism. If a genuine relationship exists in the data, it should manifest consistently across multiple risk indices, lending credibility to any pattern that does emerge.

2.4 STLFSI risk index

This results section opens with one of the most widely used risk measures produced by the Federal Reserve. Figure 2.4 below presents the impulse response function from the Federal Reserve shadow rate to the STLFSI.

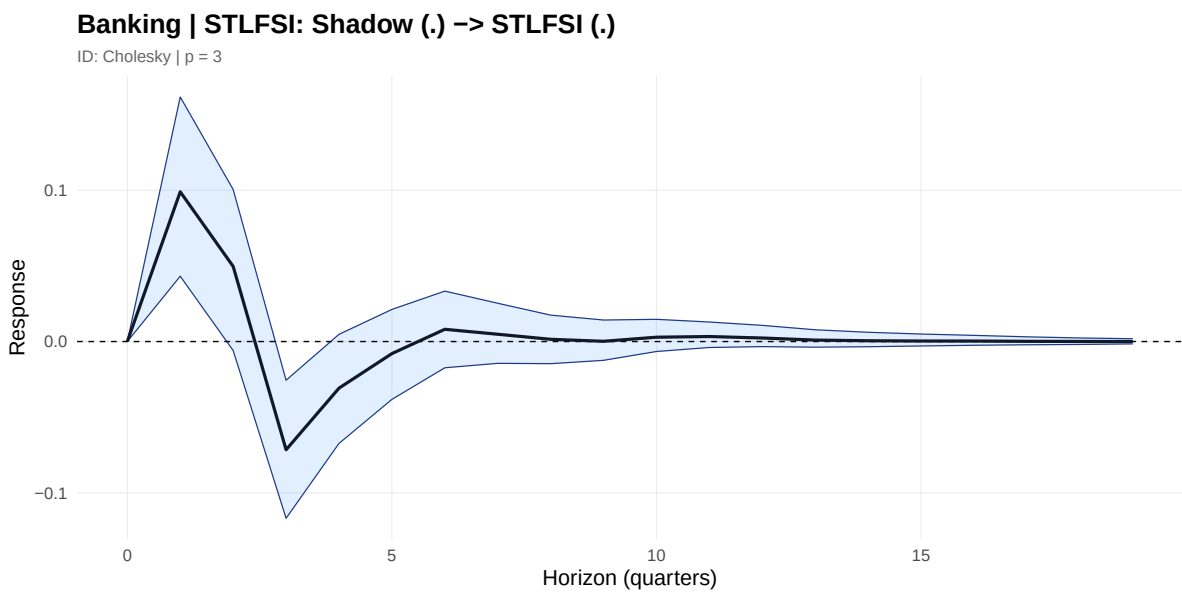


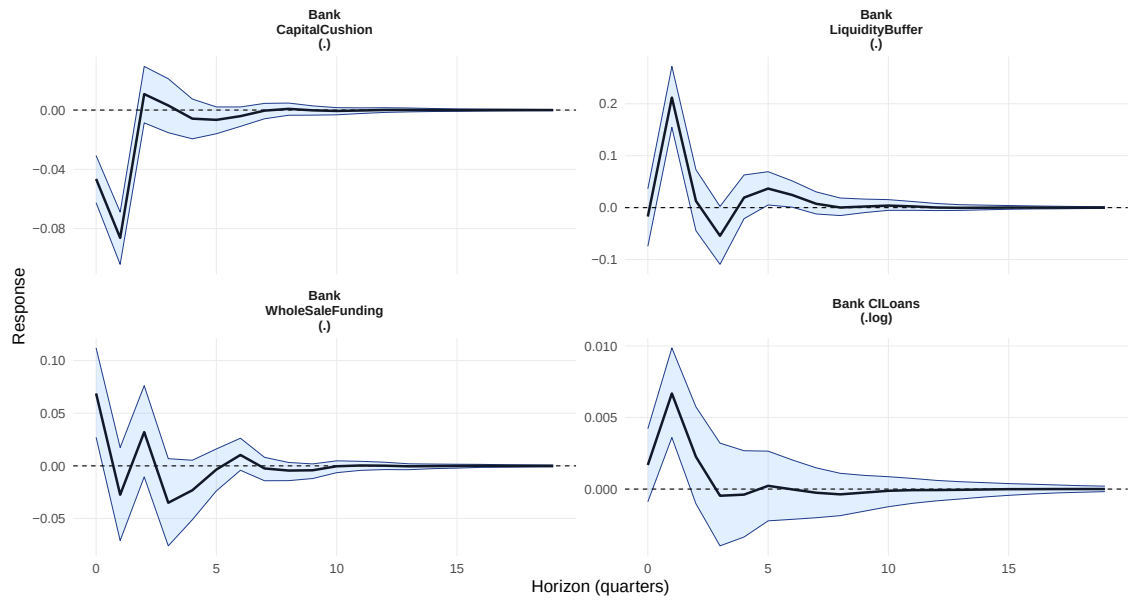
Figure 2.4: Impulse response of the STLFSI to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

When the Cholesky identification is ordered with the risk variable leading the shadow rate, the price puzzle disappears entirely. Under the alternative ordering, where the shadow rate enters the matrix as the leading variable, the price puzzle manifests as an initial negative impulse response of risk following a contractionary monetary policy shock. This negative response is subsequently followed by the dynamics presented in Figure 2.4. Once the ordering is reversed to place risk first, the price puzzle is eliminated: a rise in the shadow interest rate is instead accompanied by an immediate increase in risk, which then gives way to a dip in the third quarter before the effect of the shadow rate dissipates.

The remainder of the transmission mechanism is presented in Figure 2.5 below, which covers all subsectors of the financial sector and their respective impulse response functions.

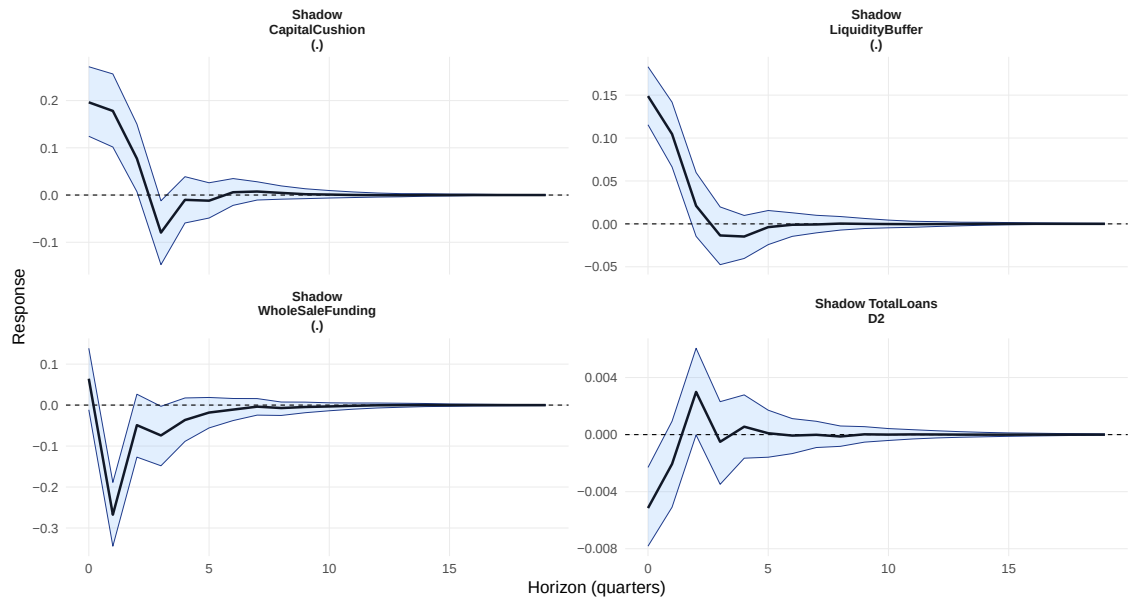
Banking | STLFSI: Responses to STLFSI (.) shock

ID: Cholesky | $p = 3$ | sector: Banking



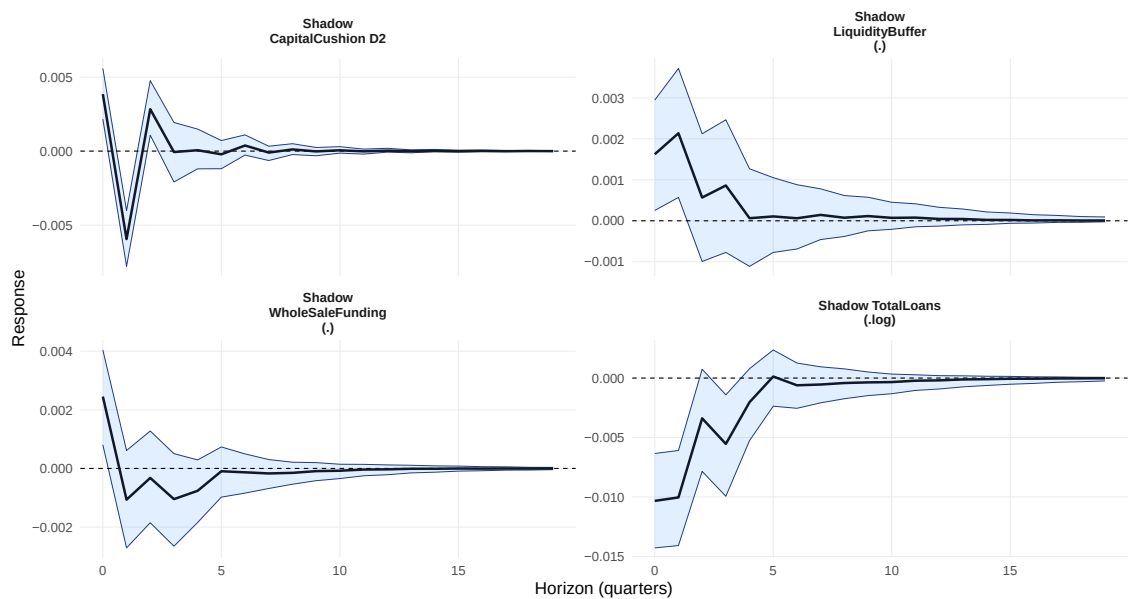
Shadow Banking (Broad) | STLFSI: Responses to STLFSI (.) shock

ID: Cholesky | $p = 3$ | sector: Shadow



Shadow Banking (Core) | STLFSI: Responses to STLFSI (.) shock

ID: Cholesky | $p = 3$ | sector: ShadowCore



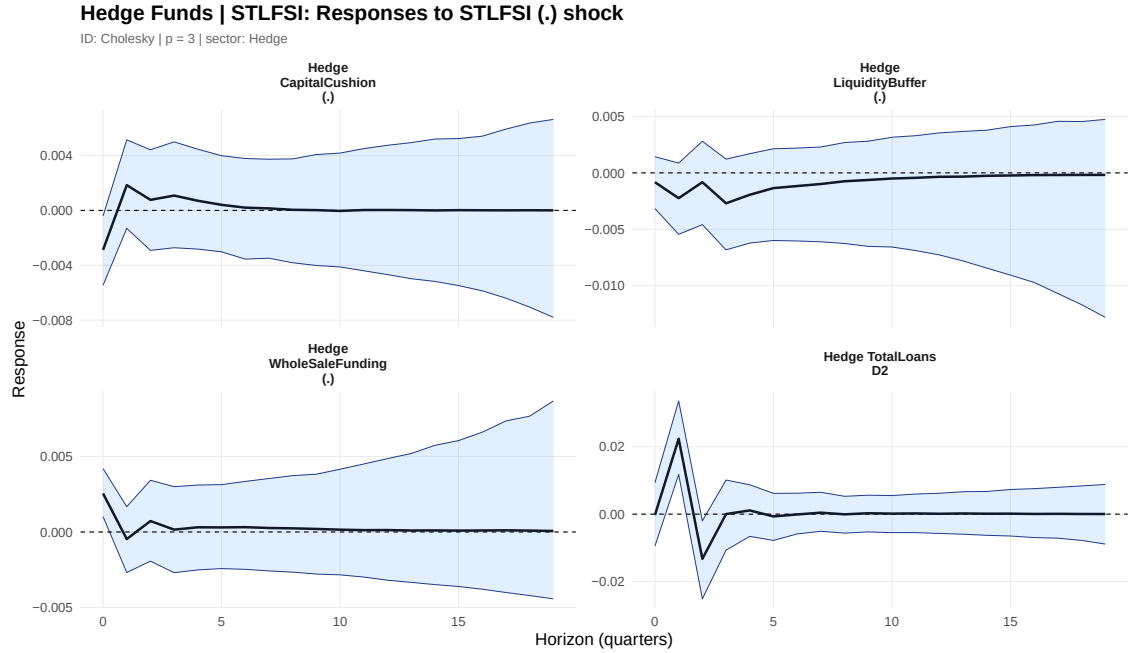


Figure 2.5: Impulse responses of sectoral variables to a STLFSI shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

In the case of broad shadow banking, the dynamics are entirely different from those observed in the traditional banking sector. Broad shadow banking rapidly attempts to deleverage, with the capital cushion fraction rising as equity accounts for a larger share of the denominator. This is again connected to wholesale funding, which decreases sharply. One behaviour that both subsectors share is a strong preference for holding more liquid assets on the balance sheet. There is a positive spike in liquid assets immediately after the risk shock propagates through the subsector, lasting roughly two quarters, consistent with almost every other shock in this transmission mechanism. With respect to total loans, broad shadow banking shows a decrease and an unwillingness to lend even at higher risk premia, which is the opposite of the response observed in the traditional banking sector.

Turning to the differences between broad and core shadow banking, several distinctions are apparent. Core shadow banking appears to more closely mimic the traditional banking sector in its leverage behaviour. Initially the sector seems to deleverage, but this is subsequently followed by rising wholesale funding and a declining capital cushion. The liquidity buffer responds positively but less strongly than in the banking and broad shadow banking cases. Total loans issued appear negative throughout, showing the largest decline among all subsectors.

The hedge fund subsector, as expected, does not react strongly to pressure from the risk side. There is a hint of increased leverage and wholesale funding at the outset, but the effect is considerably weaker and less precisely estimated compared to the other sectors. The liquidity buffer shows little discernible increase. The only well-identified reaction is in loans issued, which rises briefly a few quarters after the shock before declining by a roughly equivalent amount.

Overall, each sector displays highly heterogeneous responses to the risk shock. The theoretical explanations for this behaviour are developed further in the discussion section.

2.5 VIX risk index

The VIX index is one of the most widely used non-Fed risk indicators for monitoring the risk channel and capturing broader risk-related dynamics in financial markets. As shown in Figure 2.6, we observe a clear and consistent response following a hike in the shadow rate of the Federal Reserve: the VIX rises sharply in the initial periods and subsequently falls to negative values by the third quarter following the shock. This pattern is in line with theoretical expectations, and its clarity in the estimated impulse responses provides additional support for the validity of the identification strategy and the overall robustness of the results.

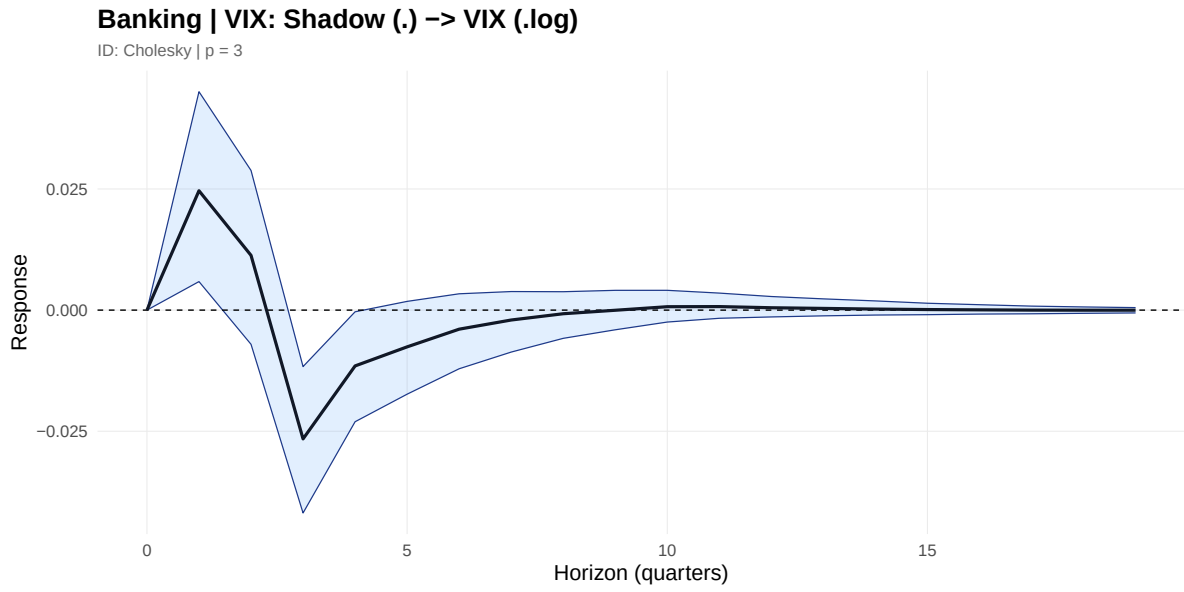
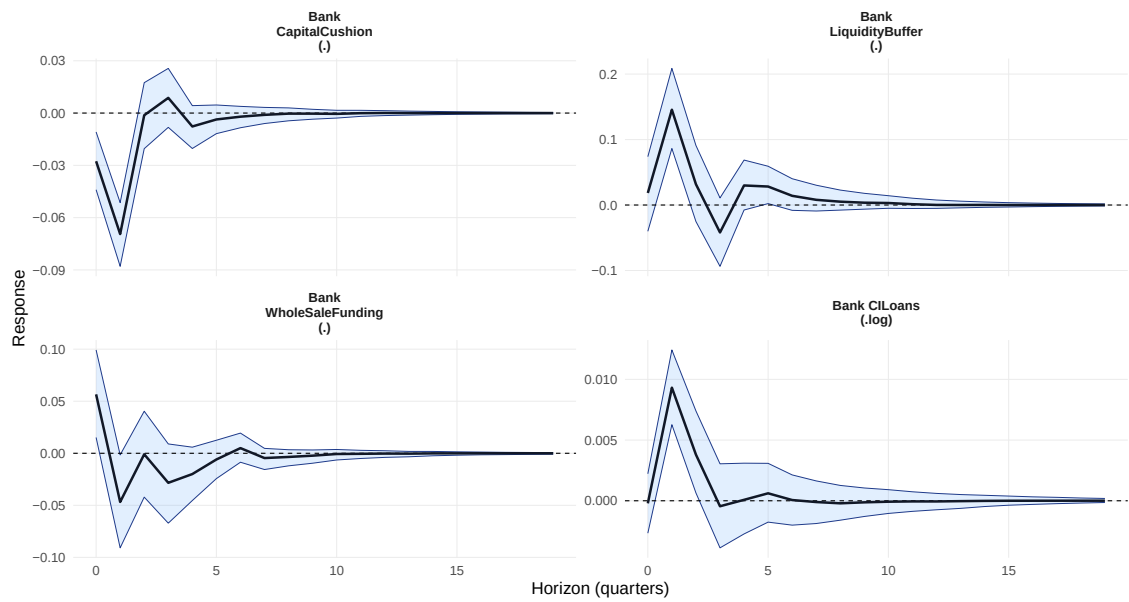


Figure 2.6: Impulse response of the VIX to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

This response to the shadow rate shock is then tracked as it propagates through the subsector variables. The layout follows the same structure as before, with all sectoral responses presented in Figure 2.7 below.

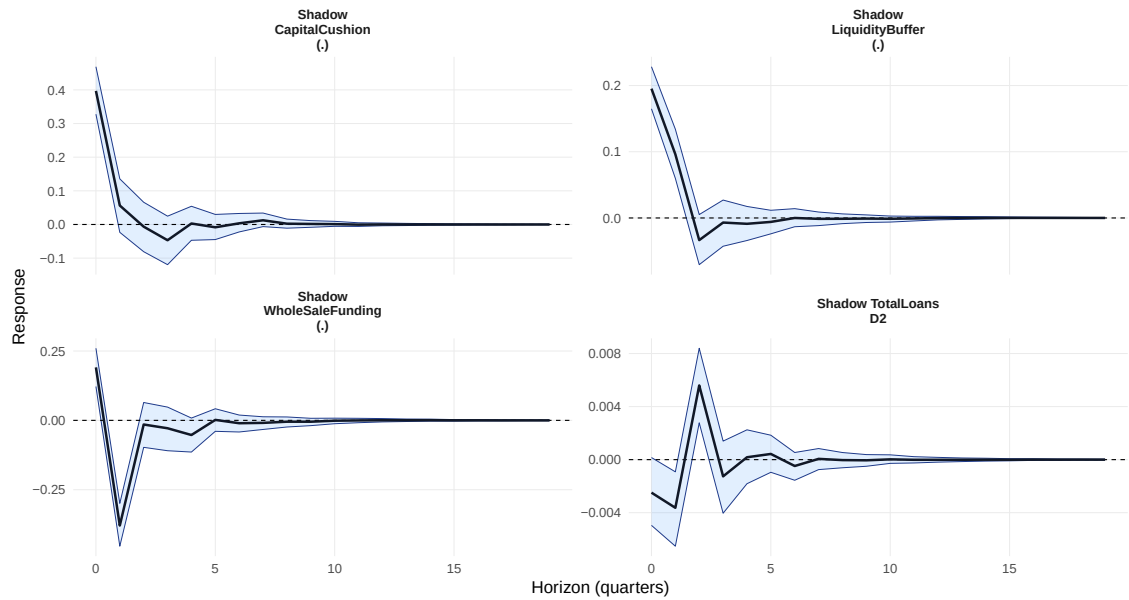
Banking | VIX: Responses to VIX (.log) shock

ID: Cholesky | $p = 3$ | sector: Banking



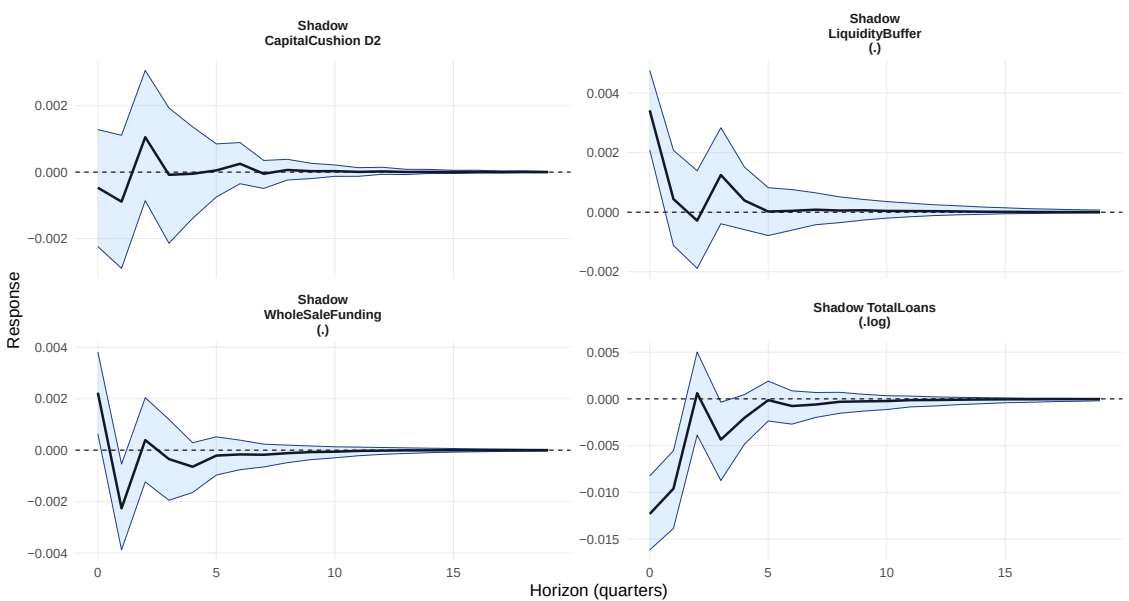
Shadow Banking (Broad) | VIX: Responses to VIX (.log) shock

ID: Cholesky | $p = 3$ | sector: Shadow



Shadow Banking (Core) | VIX: Responses to VIX (.log) shock

ID: Cholesky | $p = 3$ | sector: ShadowCore



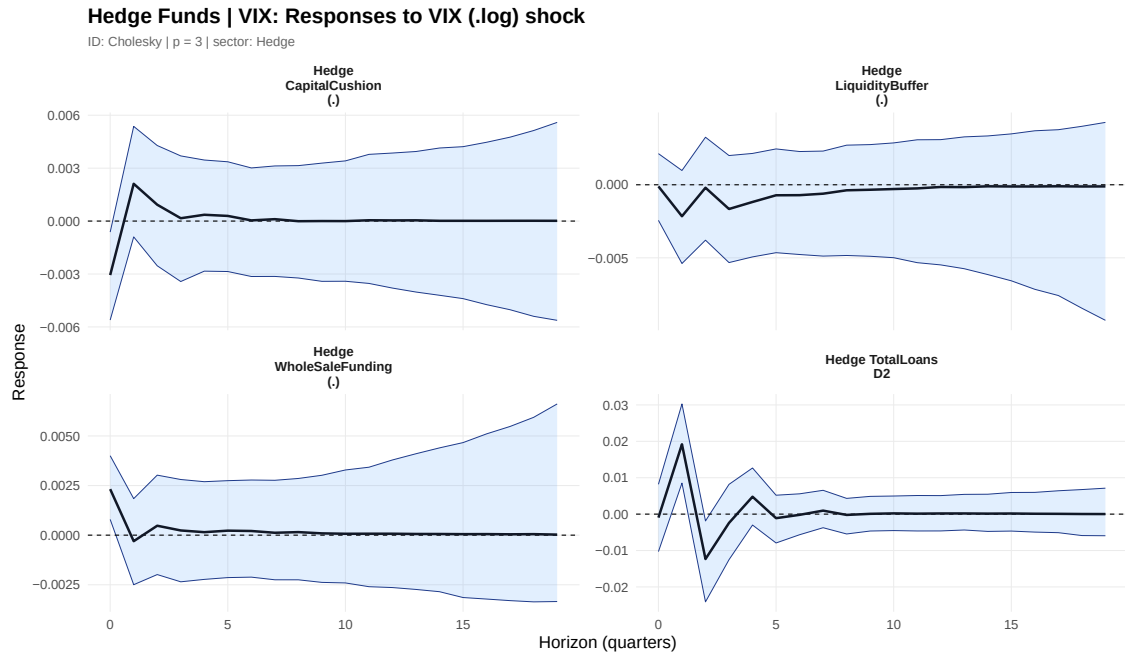


Figure 2.7: Impulse responses of sectoral variables to a VIX shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

The VIX index captures the same overall behaviour of the banking sector, with the sector increasing its leverage following the risk shock. This is reflected in both the capital cushion and wholesale funding variables. The capital cushion starts negative, meaning the fraction of total assets in the denominator is increasing, while wholesale funding rises in tandem, reinforcing the relationship. The liquidity buffer increases the most, as was the case previously, with a slight hint of a potential decline after three quarters, possibly reflecting a gradual stabilisation of risk conditions. Loans are again positive, reflecting a greater willingness to lend at higher risk premia.

Broad shadow banking exhibits a strong deleveraging reaction to the risk shock. In this case the reaction of the capital cushion is the strongest, with the equity fraction increasing sharply, which is again connected to a decline in wholesale funding accompanied by very narrow confidence intervals. Both reactions persist for roughly three quarters. Loans appear initially negative and then peak after approximately three months, with this three-quarter peak being more pronounced than in the case of the STLFSI index.

Core shadow banking in this case detects a similar but more blurred relationship compared to the previous specification. The reaction of the capital cushion is ambiguous, with very wide confidence intervals, while wholesale funding hints at a possible increase in borrowing followed by a decrease of roughly equal magnitude. The liquidity buffer reaction is positive but substantially smaller in magnitude compared to the other subsectors, consistent with the earlier results. The number of loans appears to follow a negative impulse response throughout, with a slight peak after three quarters, similarly to the broad shadow banking subsector. The hedge fund subsector shows a reaction that appears nearly identical to that observed under the STLFSI risk index, with the only significant and clearly readable response found in issued loans, which spike positively and then revert after three quarters. The early periods of the impulse response function and the wholesale funding variable also hint at a possible increase in leverage.

Overall, the reactions of the subsectors to the VIX index do not differ substantially from those

observed under the STLFSI index, which is an encouraging sign for the identification of the risk channel of monetary policy and the visible heterogeneity in subsector responses.

2.6 NSI (Shapiro) risk index

The Shapiro news sentiment index is the first fully sentiment-based index for which results are presented in this thesis. Its impulse response function is shown in Figure 2.8 below.

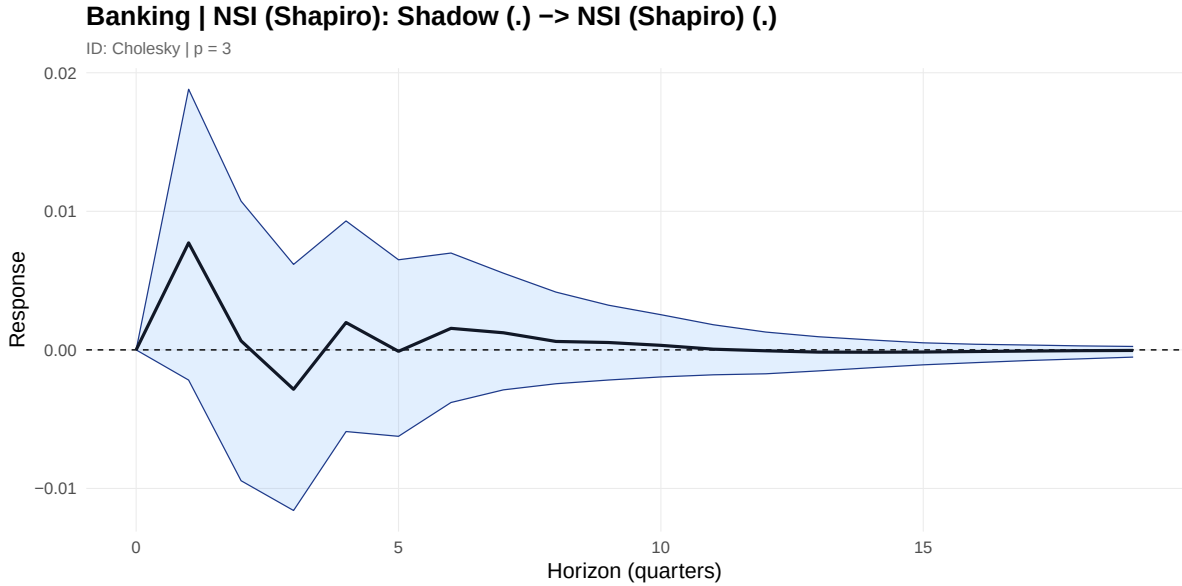
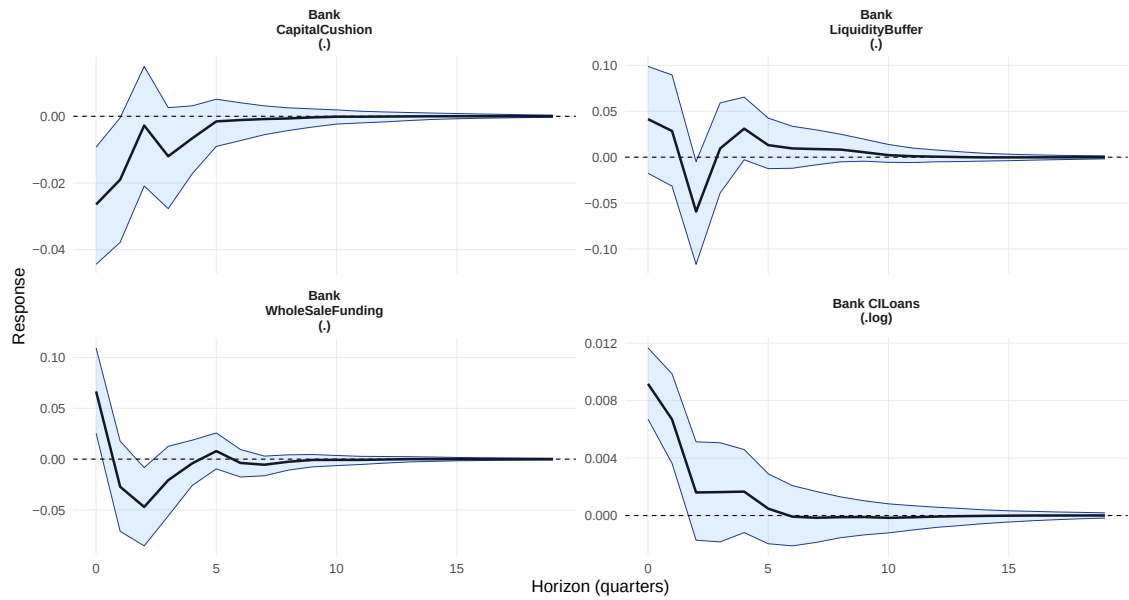


Figure 2.8: Impulse response of the NSI (Shapiro) to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

The confidence intervals in this impulse response function are wide, but there is a hint of a positive spike that dissipates after two quarters, as the majority of draws are positive in this period, pushing the median of the interval above zero. The responses of the sectoral variables to the sentiment index are shown in Figure 2.9 below.

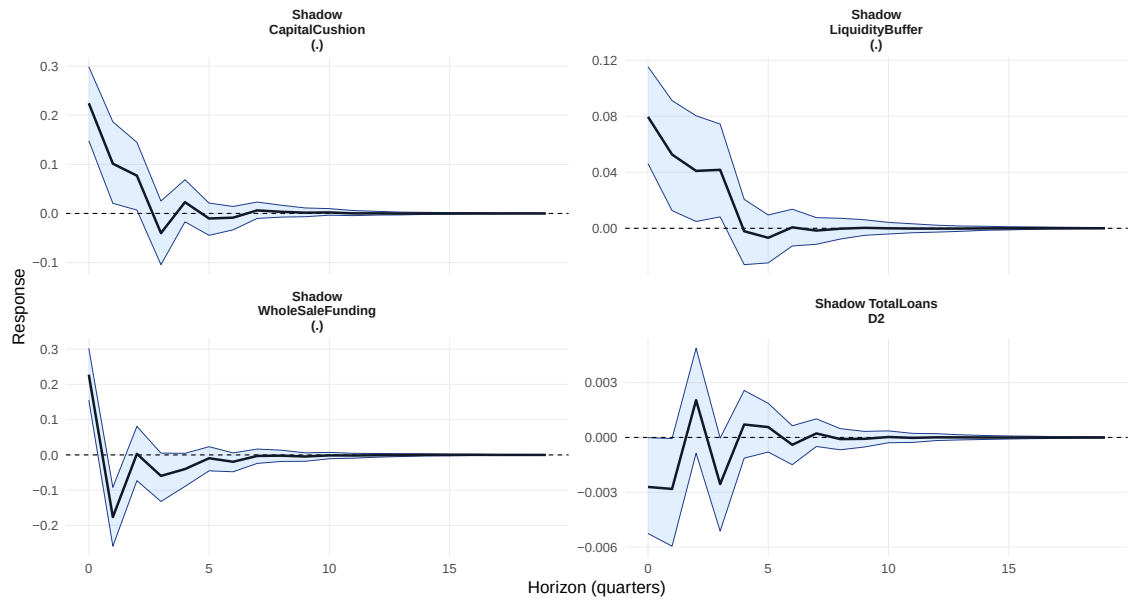
Banking | NSI (Shapiro): Responses to NSI (Shapiro) (.) shock

ID: Cholesky | p = 3 | sector: Banking



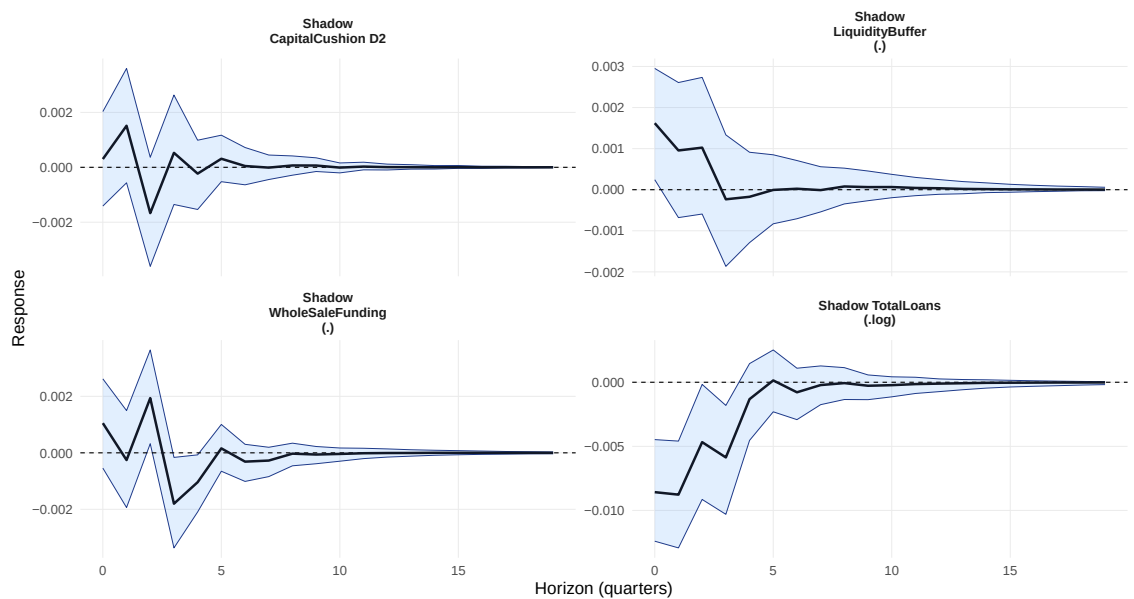
Shadow Banking (Broad) | NSI (Shapiro): Responses to NSI (Shapiro) (.) shock

ID: Cholesky | p = 3 | sector: Shadow



Shadow Banking (Core) | NSI (Shapiro): Responses to NSI (Shapiro) (.) shock

ID: Cholesky | p = 3 | sector: ShadowCore



Hedge Funds | NSI (Shapiro): Responses to NSI (Shapiro) (.) shock

ID: Cholesky | $p = 3$ | sector: Hedge

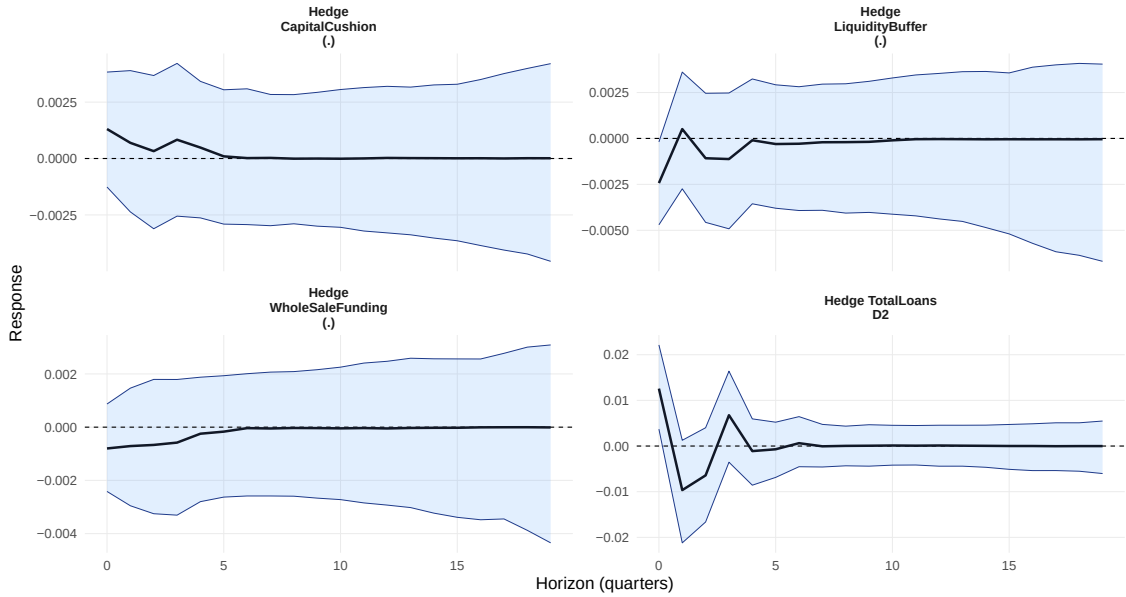


Figure 2.9: Impulse responses of sectoral variables to a NSI (Shapiro) shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

With the difference in construction of the risk index, the results for the banking sector are slightly different from those observed previously. The responses are similar in direction but carry a subtle nuance introduced by the different input source. The capital cushion still decreases, accompanied by a rise in wholesale funding that ends by the third quarter for both fractions. The most notable change can be seen in the reaction of the liquidity buffer, which appears to start positive but then turns negative after three quarters, dipping before returning to the baseline. This may be caused by a different lag structure in the observed shock, with the possibility that the sentiment index better captures the tail end of the shock, or reflects a specific feature of how the shock transmits to the liquidity buffer. Bank loans are positive from the outset, consistent with the two previous indices.

For broad shadow banking, the results are largely consistent with those observed before. The subsector is deleveraging, with the equity fraction increasing on the balance sheet, though in this case wholesale funding drops with some lag. The liquidity buffer shows a clear positive spike lasting nearly four quarters. The behaviour of total loans is more puzzling: they appear to decrease following the shock, then increase before reverting to the baseline. The confidence intervals suggest that the initial impact is negative, with the majority of draws falling below zero.

The sentiment index does not perform as well in explaining the reactions of the core shadow banking subsector. The identification for both the capital cushion and wholesale funding is blurred, with very wide confidence intervals. There is a hint of increasing leverage around the third quarter, suggested by the capital cushion and wholesale funding trends edging slightly above zero, but this remains inconclusive. The only clear, if weak, responses are found in the liquidity buffer and total loans issued. The liquidity buffer increases slightly after the shock but the reaction dissipates quickly, while total loans are negative from the outset and this reaction persists for quite a few periods, only stabilising after approximately five quarters. The reaction of the hedge fund subsector is even more ambiguous than before, with only a faint hint of issued loans increasing after the risk shock.

Overall, the sentiment index brings a different perspective to the propagation of the risk shock across

subsectors. The relationships are sometimes different compared to those identified using traditional time series based risk indices, and the reasons for these differences will be discussed further in the discussion section.

2.7 Sentiment FinBERT risk index

Results from the Shapiro sentiment index proven to be quite similar to the time series risk indices, the question is how the sentiment index created from scratch can compare to them. This is answered in this results subsection when the mine sentiment risk index created with FinBERT family of models is presented.

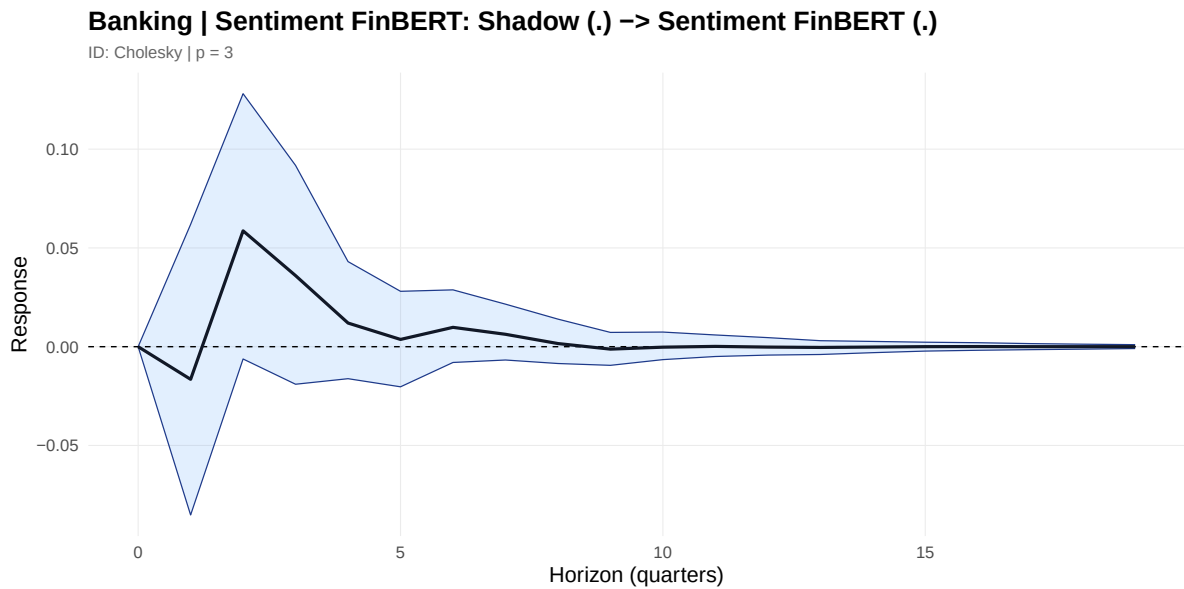
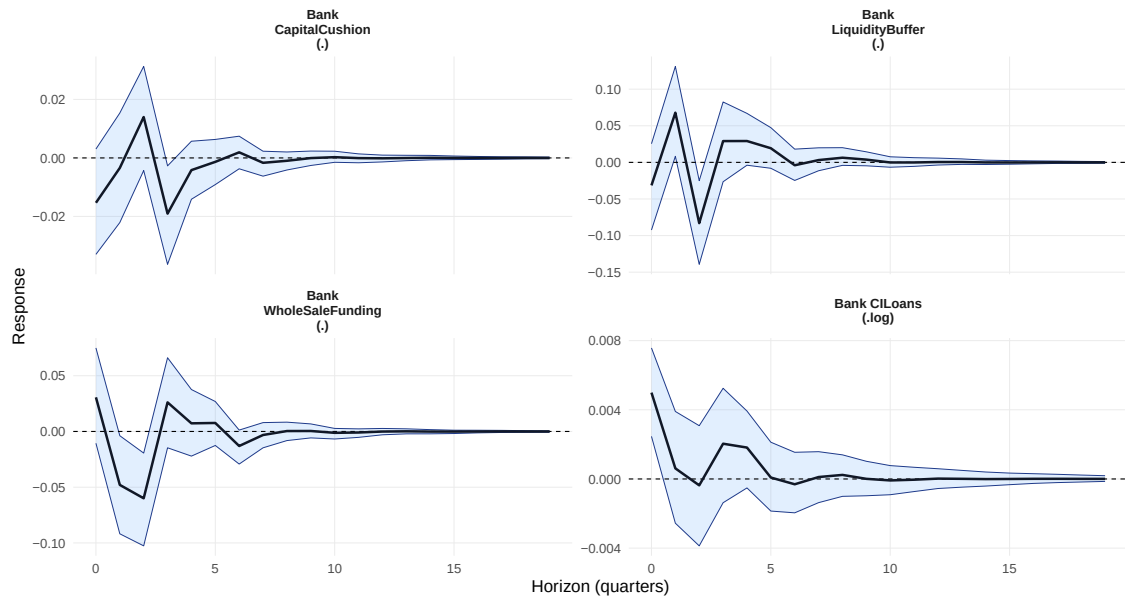


Figure 2.10: Impulse response of the Sentiment FinBERT index to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

Impulse response function coming from shadow rate to the FinBERT risk index can be seen in the figure 2.10. In similar fashion as with the Shapiro sentiment index it is blurred and the confidence intervals are only hinting on the increase in the third quarter. Rest of the sectoral impulse response function can be seen below in the figure 2.11.

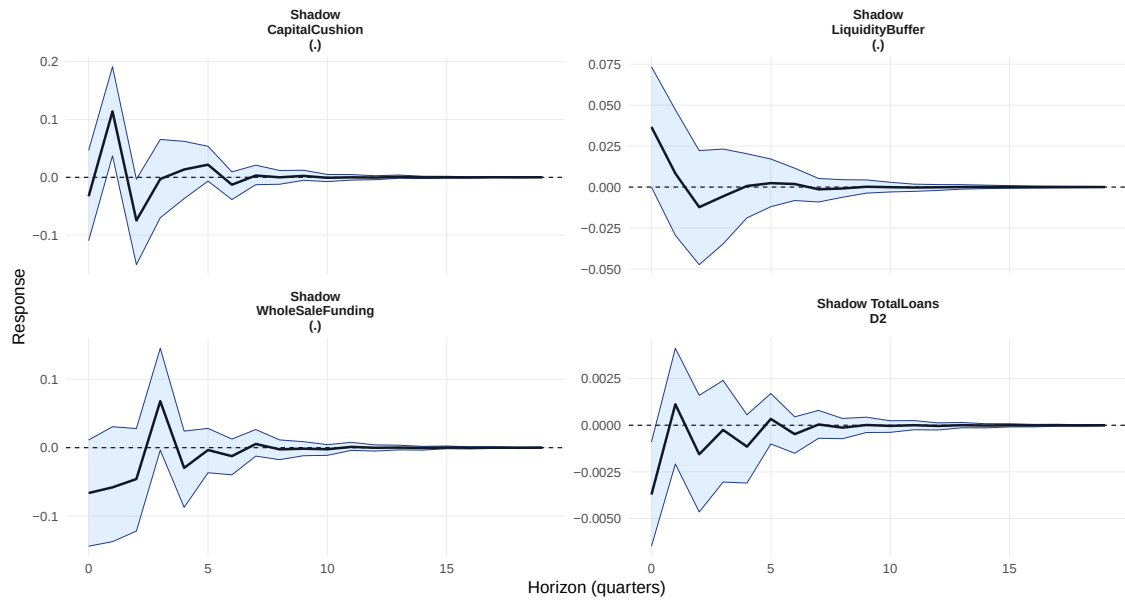
Banking | Sentiment FinBERT: Responses to Sentiment FinBERT (.) shock

ID: Cholesky | $p = 3$ | sector: Banking



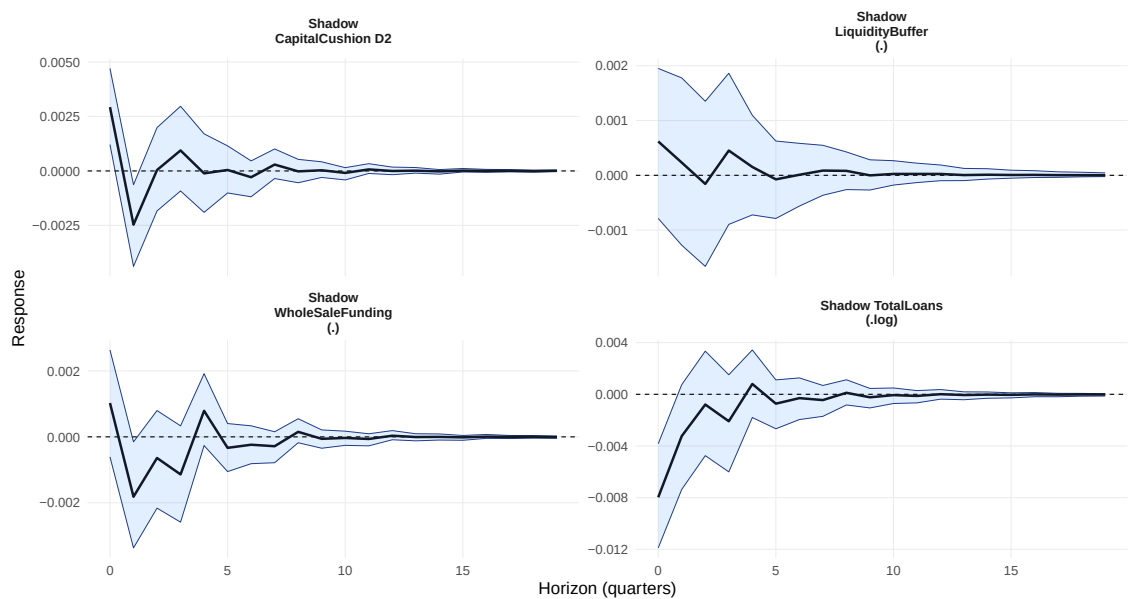
Shadow Banking (Broad) | Sentiment FinBERT: Responses to Sentiment FinBERT (.) shock

ID: Cholesky | $p = 3$ | sector: Shadow



Shadow Banking (Core) | Sentiment FinBERT: Responses to Sentiment FinBERT (.) shock

ID: Cholesky | $p = 3$ | sector: ShadowCore



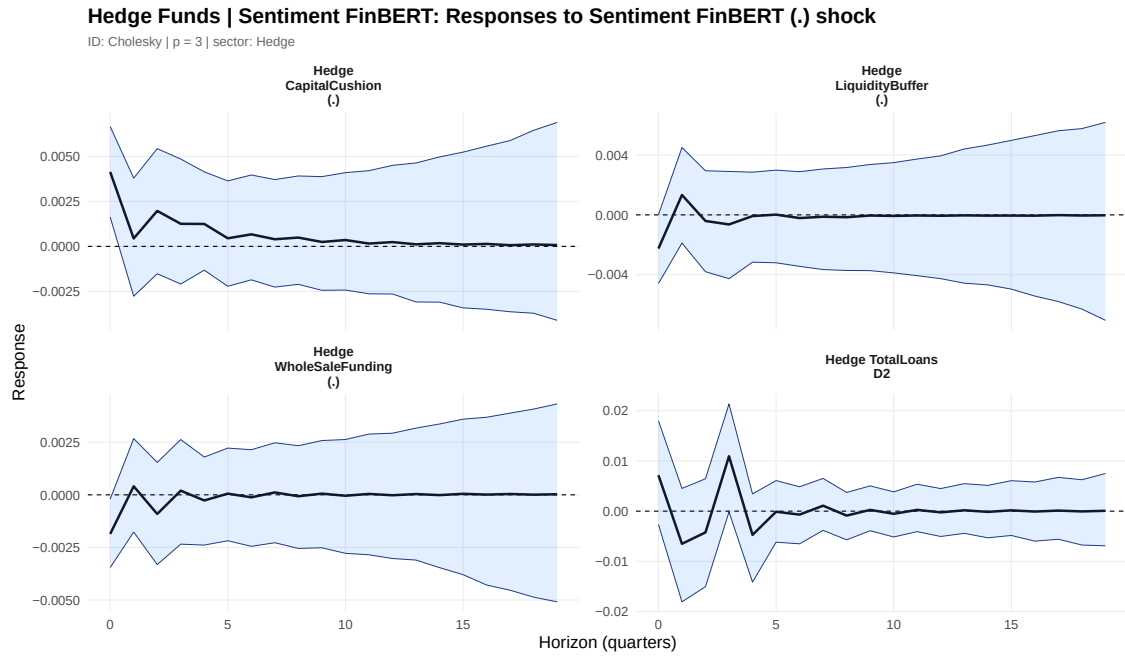


Figure 2.11: Impulse responses of sectoral variables to a Sentiment FinBERT shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

The specific details of the impulse response functions appear less important here, as the model struggles with identification of the relationships that were confirmed by the three previous risk indices. This lack of continuity with the present risk index can be primarily attributed to the volatility and instability in the index itself. The reader may recall that this is the more stable of the two sentiment indices compared to the VADER-based one. This suggests that in order to clearly leverage the usefulness of the sentiment signal, which was visible in the Shapiro results, the sentiment signal needs to be cleanly extracted and smoothed correctly across different economic regimes. Among the impulse response functions presented, some resemble the signals observed with the previous risk indices, but the majority are blurred versions of those earlier results.

The one notable exception is the hedge fund subsector, which appears to show a clearer initial reaction to the spike in the sentiment index across all sectoral variables.

2.8 Custom Index

In this section, I present results for my custom sentiment index, which is purely time series based. It serves as a test of how well this simple two-stage principal component risk index performs relative to the traditional and experimental risk indices.

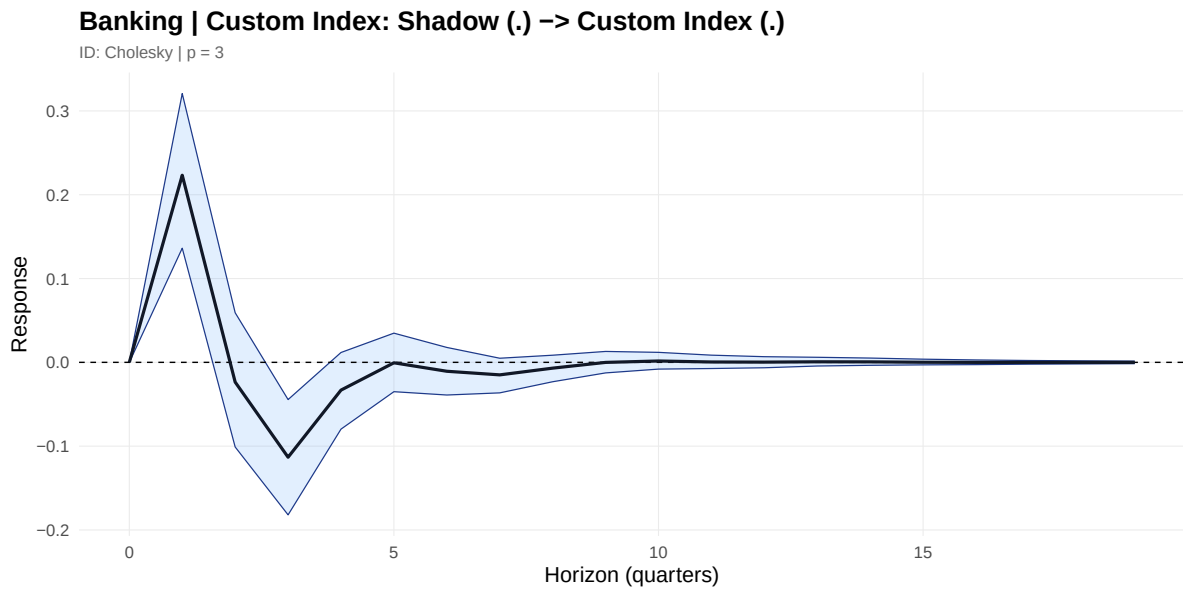
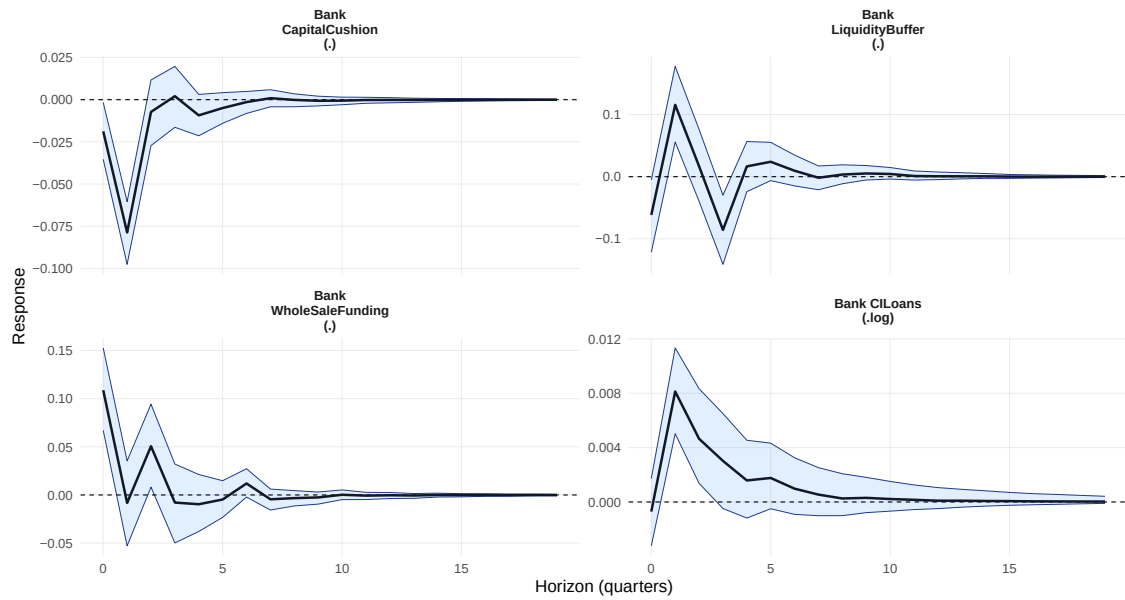


Figure 2.12: Impulse response of the Custom Index to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

The impulse response function of the index to the Federal Reserve shadow rate is shown in Figure 2.12 above. There is a classic reaction consistent with what has been observed before, with the index spiking following an increase in the shadow rate and then declining after three to four quarters. The initial identification appears to perform well compared to the previous risk indices, which is an encouraging sign for the validity of the custom index. The remaining sectoral impulse response functions are shown in Figure 2.13 below.

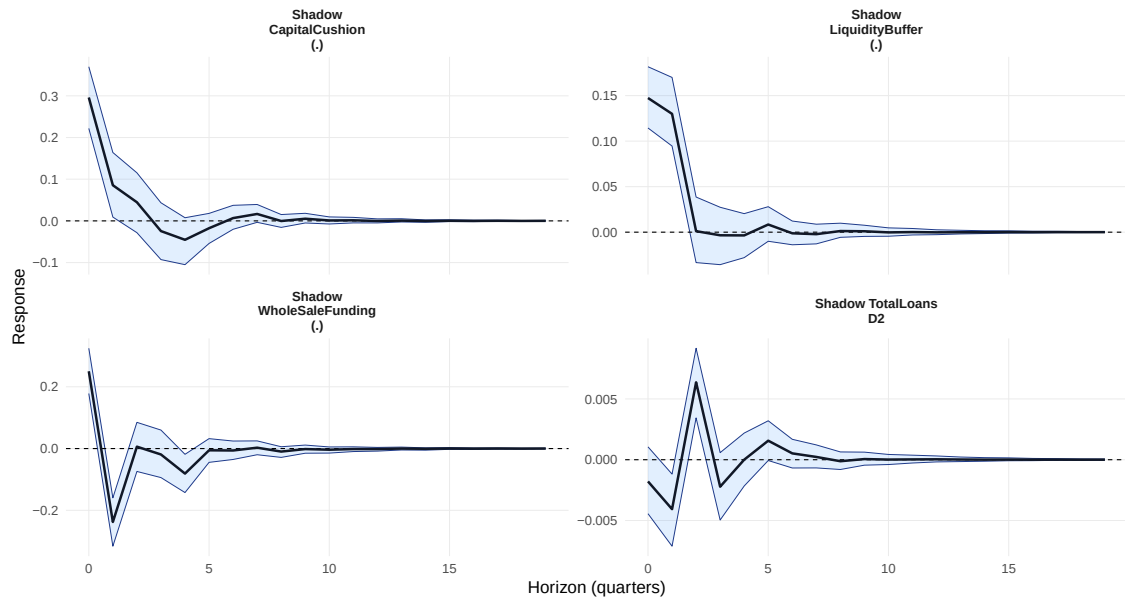
Banking | Custom Index: Responses to Custom Index (.) shock

ID: Cholesky | $p = 3$ | sector: Banking



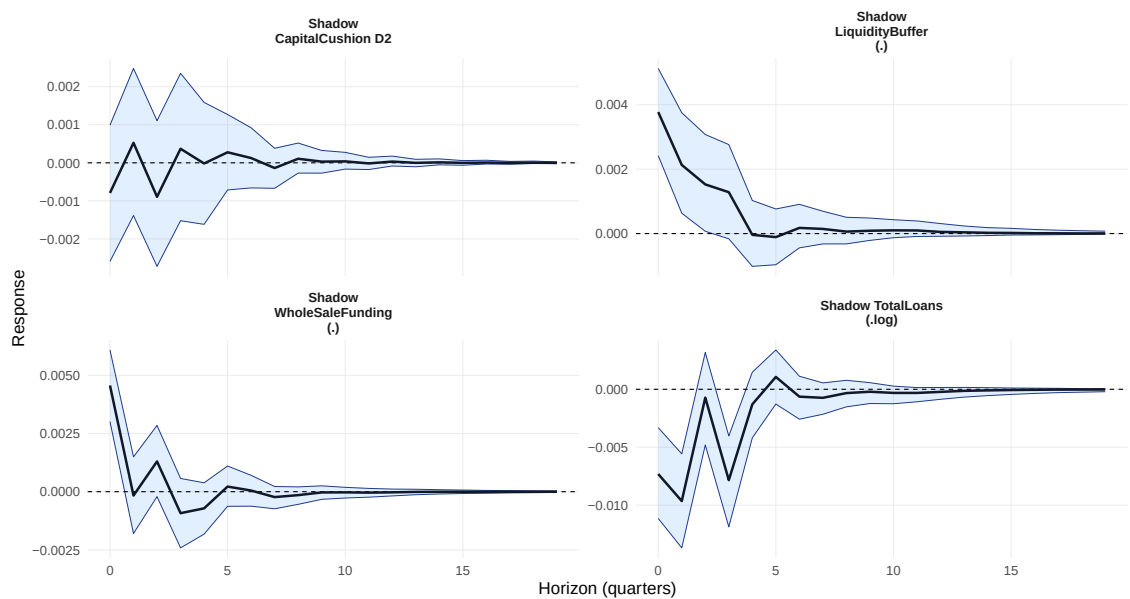
Shadow Banking (Broad) | Custom Index: Responses to Custom Index (.) shock

ID: Cholesky | $p = 3$ | sector: Shadow



Shadow Banking (Core) | Custom Index: Responses to Custom Index (.) shock

ID: Cholesky | $p = 3$ | sector: ShadowCore



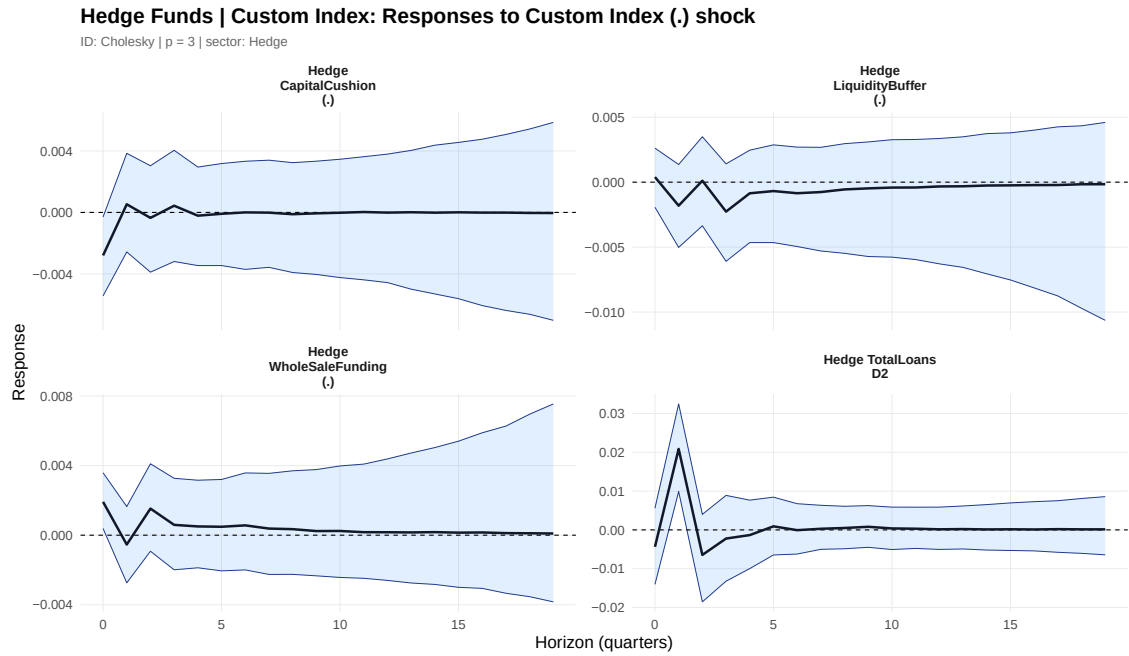


Figure 2.13: Impulse responses of sectoral variables to a Custom Index shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

Based on these results, the two-stage PCA risk index built purely on time series data without sentiment appears to be successful in identifying risk dynamics. This is most visible in the banking subsector and the broad shadow banking subsector. Starting with the banking sector, leverage increases and peaks by the third quarter, reflected in both the capital cushion and wholesale funding. This three-quarter timing appears consistently across the impulse response functions and is most likely connected to the reversion of risk following the shock to the Federal Reserve shadow rate. This is most visible in the liquidity buffer and total loans. The liquidity buffer peaks but then falls from the third quarter onward as risk decreases, with the banking sector responding to the decline in risk conditions and liquid assets becoming less necessary as the risk premia they carry diminishes. Total loans show a clean positive spike following the risk shock lasting roughly three quarters. Notably, this is the first instance where we observe a clear and statistically significant reversion in the liquidity buffer after the three-quarter mark. In previous risk indices this reversion was only hinted at through the confidence intervals, and this finding will be discussed further in the next section.

For broad shadow banking, the situation is similar to that observed with the benchmark indices. There is a strong deleveraging effect in the first three quarters accompanied by a decline in wholesale funding. The liquidity buffer remains positive until the third quarter, at which point the effect dissipates. The three-quarter pattern is also visible in total loans issued by the subsector, which spike around the third quarter as risk begins to decline and conditions normalise.

Core shadow banking behaves approximately as before with the benchmark indices. The main issue is that the capital cushion remains ambiguous due to wide confidence intervals. Wholesale funding is increasing, hinting at a rise in leverage that is not clearly visible in the capital cushion. The liquidity buffer is positive and consistent across all risk indices. Total loans appear initially negative, followed by a slight spike around the third quarter, but the overall effect thereafter remains negative and supports the previous results.

For the hedge fund subsector, the narrative is similar to the previous specification. There is a faint

hint of an initial increase in leverage, visible in both the capital cushion and wholesale funding. The clearest impulse response function is found in loan issuance, which spikes positively in the first to second quarter and then reverts, indicating potential capitalisation on risk premia.

Overall, these results represent a success for the custom risk index and demonstrate that a simple two-stage PCA approach can capture broad risk dynamics and, in some impulse response functions, possibly trace the complete evolution of risk propagation through the financial system.

2.9 Custom Mix

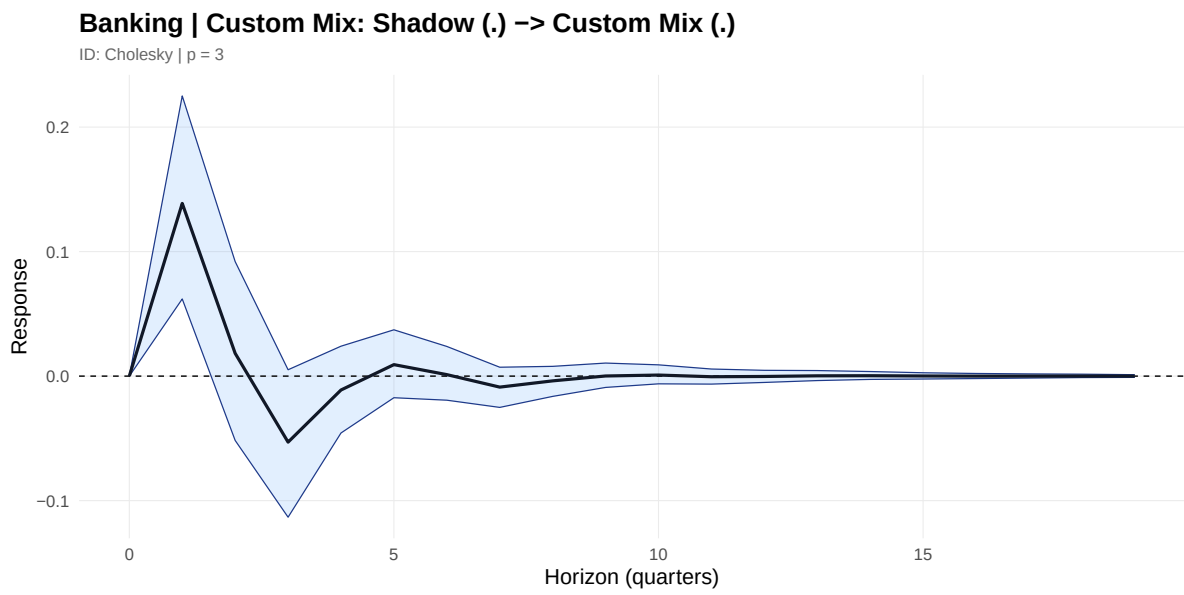
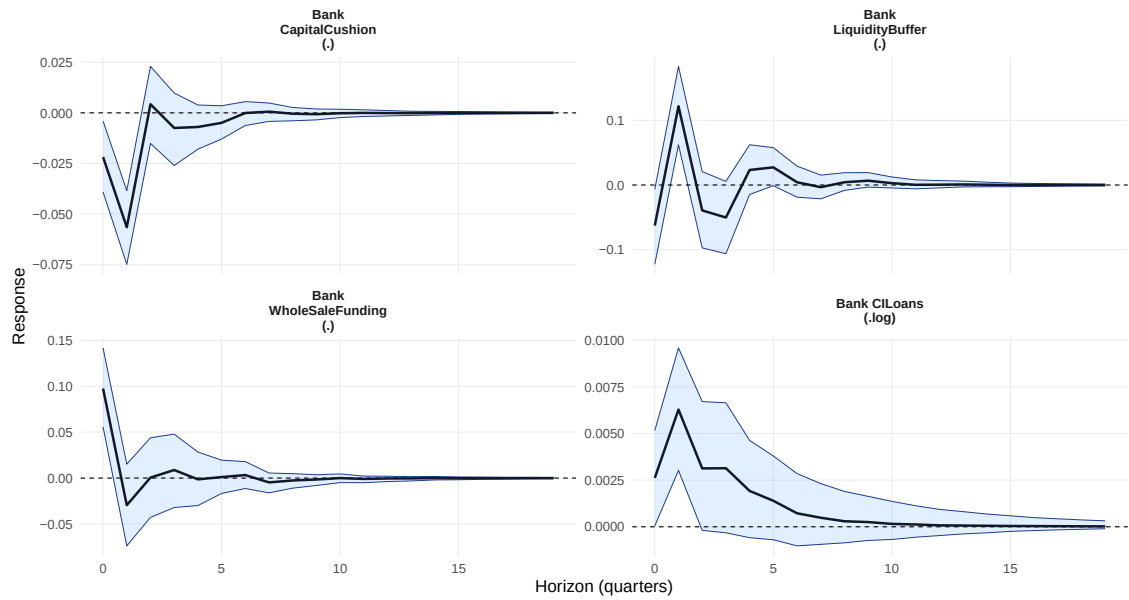


Figure 2.14: Impulse response of the Custom Mix index to a shadow rate shock. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

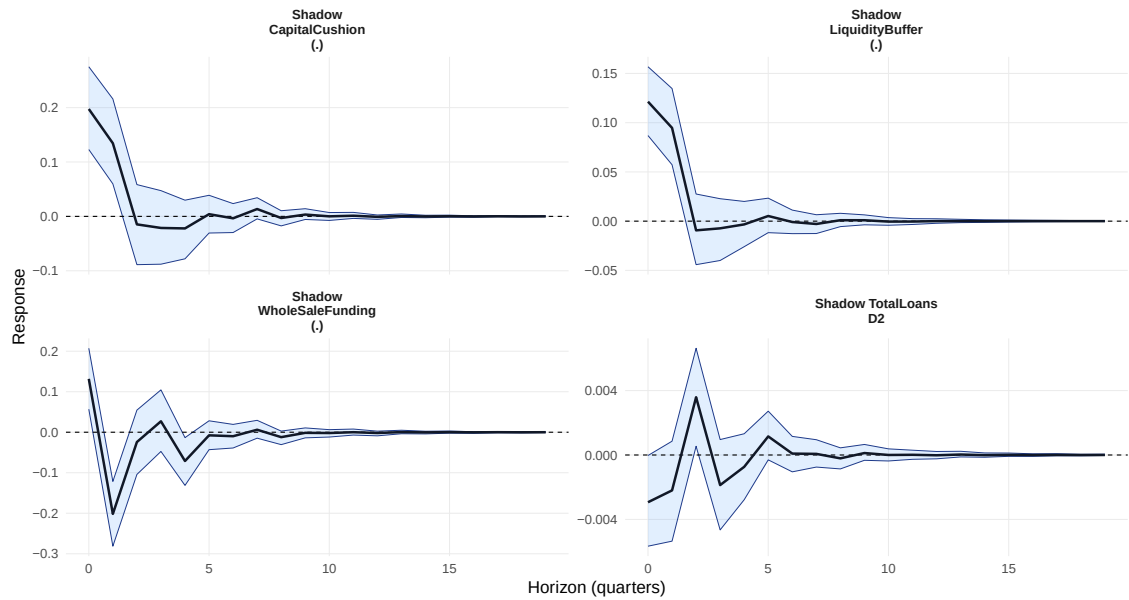
Banking | Custom Mix: Responses to Custom Mix (.) shock

ID: Cholesky | $p = 3$ | sector: Banking



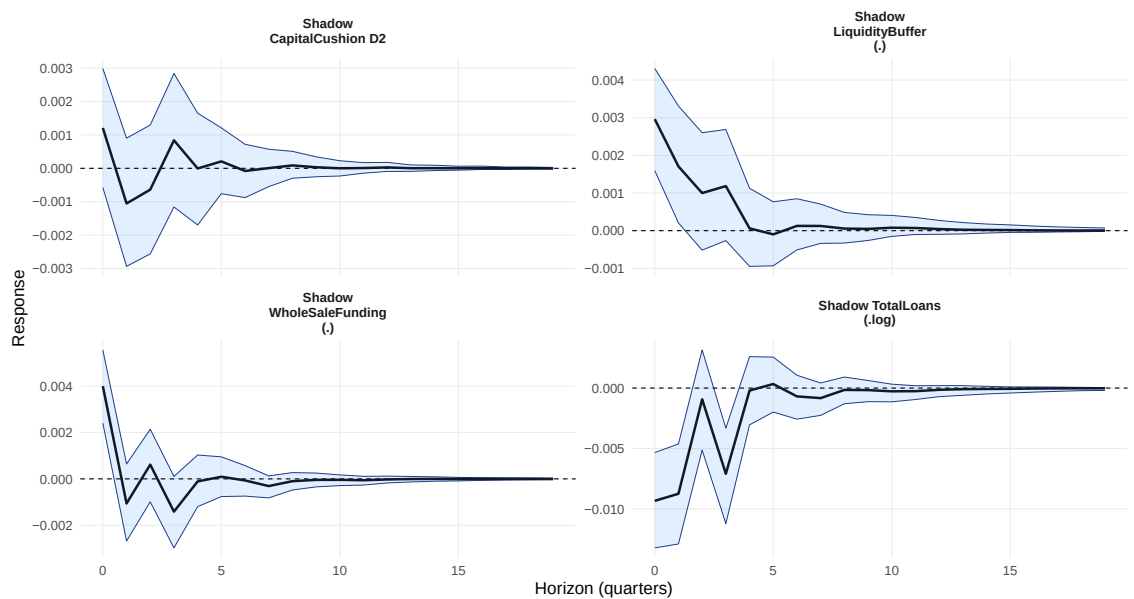
Shadow Banking (Broad) | Custom Mix: Responses to Custom Mix (.) shock

ID: Cholesky | $p = 3$ | sector: Shadow



Shadow Banking (Core) | Custom Mix: Responses to Custom Mix (.) shock

ID: Cholesky | $p = 3$ | sector: ShadowCore



Hedge Funds | Custom Mix: Responses to Custom Mix (.) shock

ID: Cholesky | $p = 3$ | sector: Hedge

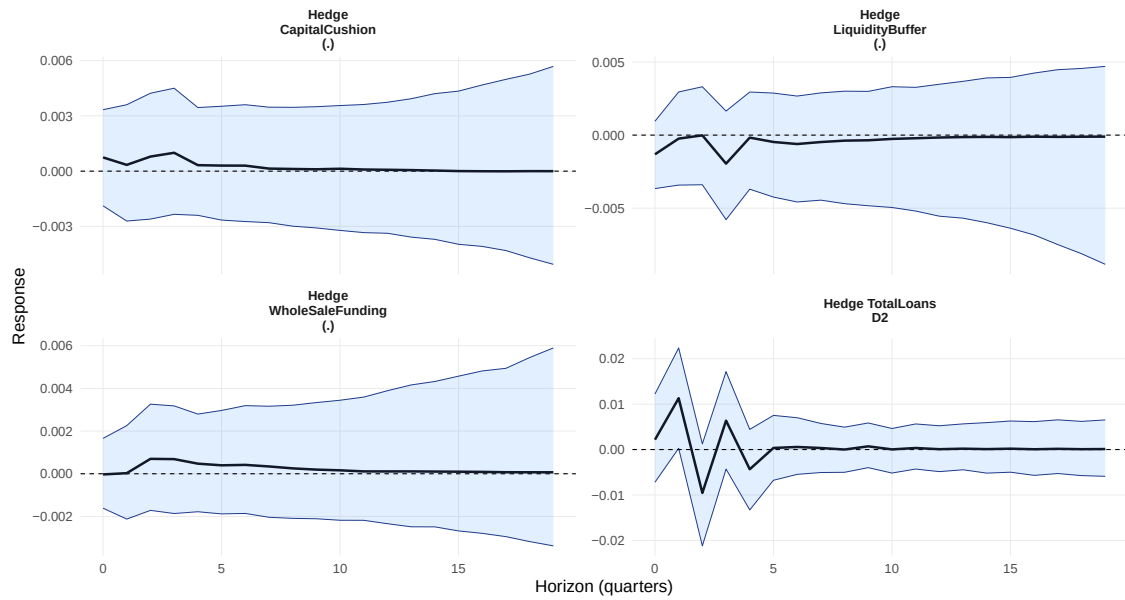


Figure 2.15: Impulse responses of sectoral variables to a Custom Mix shock. Top: Banking sector. Second: Shadow banking sector (broad). Third: Shadow banking sector (core). Bottom: Hedge fund sector. Shaded bands represent 68% and 90% posterior credible intervals. Identification via Cholesky decomposition, $p = 3$ lags.

Chapter 3

Conclusion

[Write conclusion]